Puberty and emotions study model outputs

H1 Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3054924 0.05547769 0.1908525 0.4070574
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.62	-0.62	[-0.88, -0.36]	100%
b_pds_total_s	0.21	0.21	[-0.06, 0.46]	94.07%
b_adhd_statusY	0.98	0.98	[0.65, 1.30]	100%
b_age_cont_s	-2.06e-03	-2.37e-03	[-0.21, 0.21]	50.90%
b_sed_percentile_s	0.09	0.09	[-0.06, 0.24]	86.88%
b_pds_total_s:adhd_statusY	-0.06	-0.06	[-0.37, 0.25]	64.91%

Parameter	% in ROPE	meaningful
b_Intercept	5.00e-03%	TRUE
b_pds_total_s	19.81%	FALSE
b_adhd_statusY	0%	TRUE
b_age_cont_s	66.27%	FALSE
b_sed_percentile_s	55.54%	FALSE
b_pds_total_s:adhd_statusY	44.52%	FALSE

H1 Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

```
# A tibble: 13 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00  122125.   92119.
 2 b_pds_total_s               1.00   79798.   85583.
 3 b_adhd_statusY              1.00  121409.   90782.
 4 b_age_cont_s                1.00   99308.   90006.
 5 b_sed_percentile_s          1.00   55737.   90922.
 6 b_pds_total_s:adhd_statusY  1.00   95658.   92211.
 7 sigma                       1.00  128047.   91257.
 8 Intercept                   1.00  136810.   91911.
 9 lprior                      1.00  127778.   91002.
10 lprior_b                    1.00  127025.   92853.
11 lprior_intercept            1.00   64620.   74024.
12 lprior_sigma                1.00  128047.   91257.
13 lp__                        1.03    2394.    3097.


=== Divergent Transitions ===

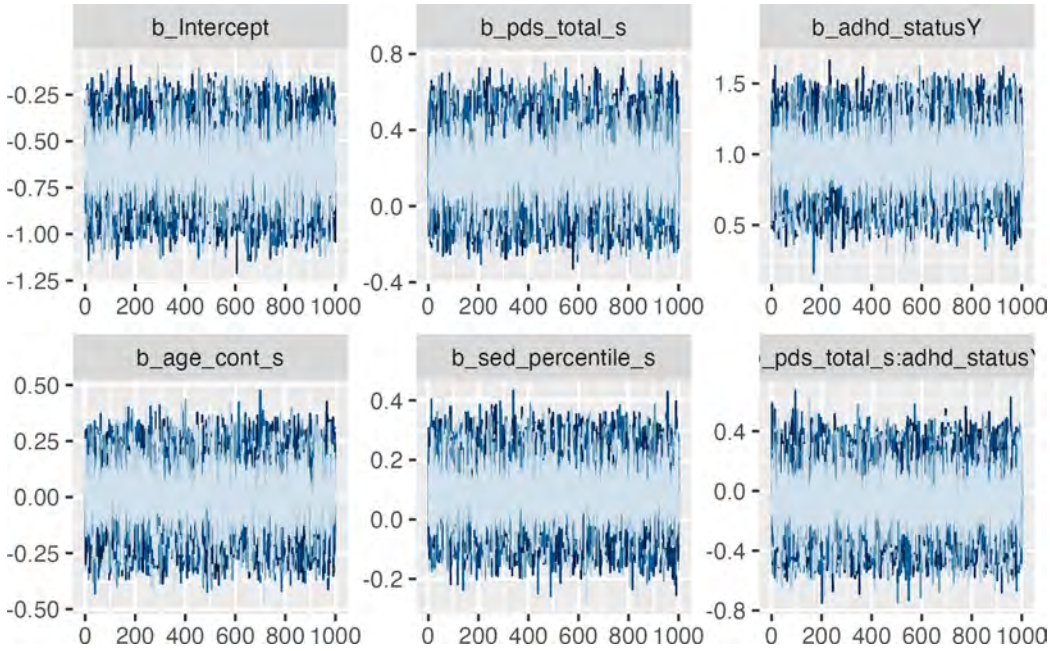
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== R-hat & ESS for lp ===

      rhat           ess_bulk       ess_tail
 Min.   :0.9997   Min.   :1583   Min.   :2137
 1st Qu.:1.0006   1st Qu.:1832   1st Qu.:2438
 Median :1.0010   Median :1904   Median :2673
 Mean   :1.0013   Mean   :1883   Mean   :2610
 3rd Qu.:1.0019   3rd Qu.:1990   3rd Qu.:2790
 Max.   :1.0035   Max.   :2069   Max.   :2922


=== Trace Plots ===
```



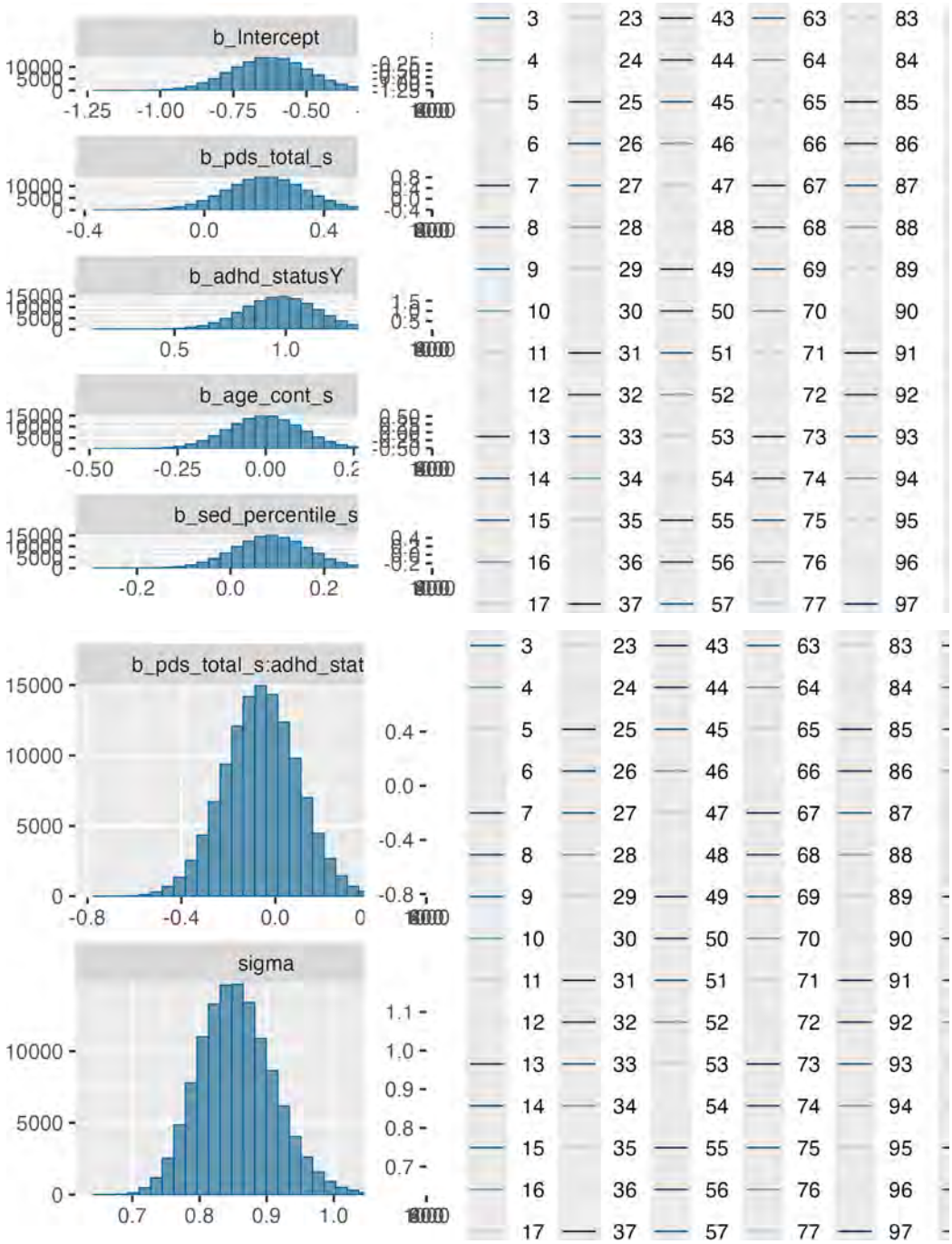

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                     coef group resp dpar nlpar lb ub
   normal(0, 1)         b
   normal(0, 1)         b             adhd_statusY
   normal(0, 1)         b               age_cont_s
   normal(0, 1)         b              pds_total_s
   normal(0, 1)         b pds_total_s:adhd_statusY
   normal(0, 1)         b         sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                    0
       tag       source
         b         user
         b (vectorized)
         b (vectorized)
         b (vectorized)
         b (vectorized)
         b (vectorized)
 intercept         user
     sigma         user
```

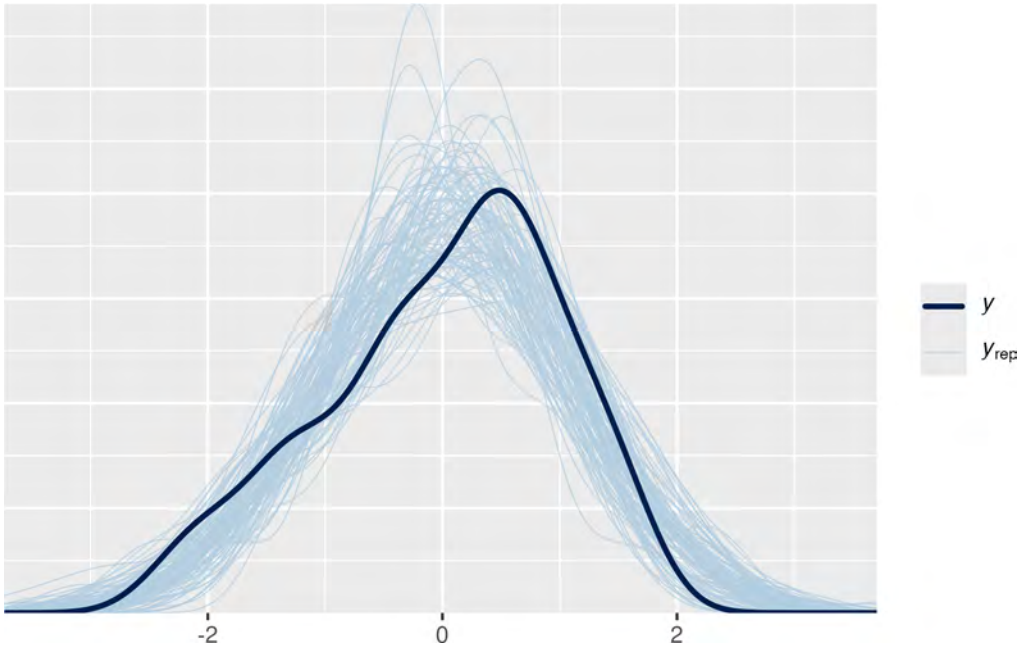
```
=== Posterior Predictive Checks ===

Density plots for each parameter.
```



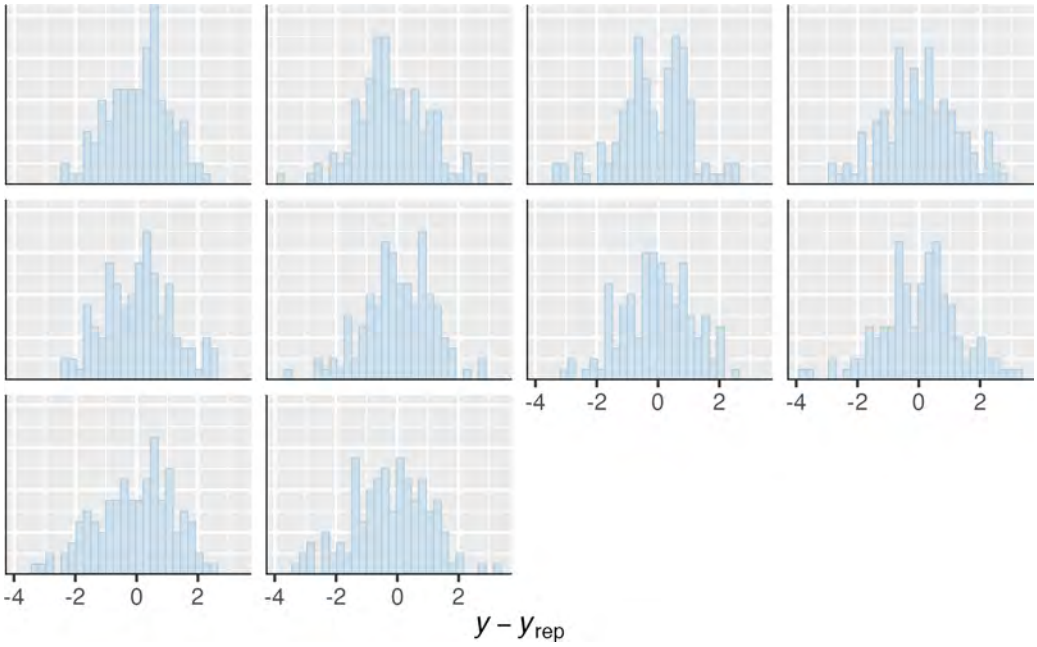
```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.
```



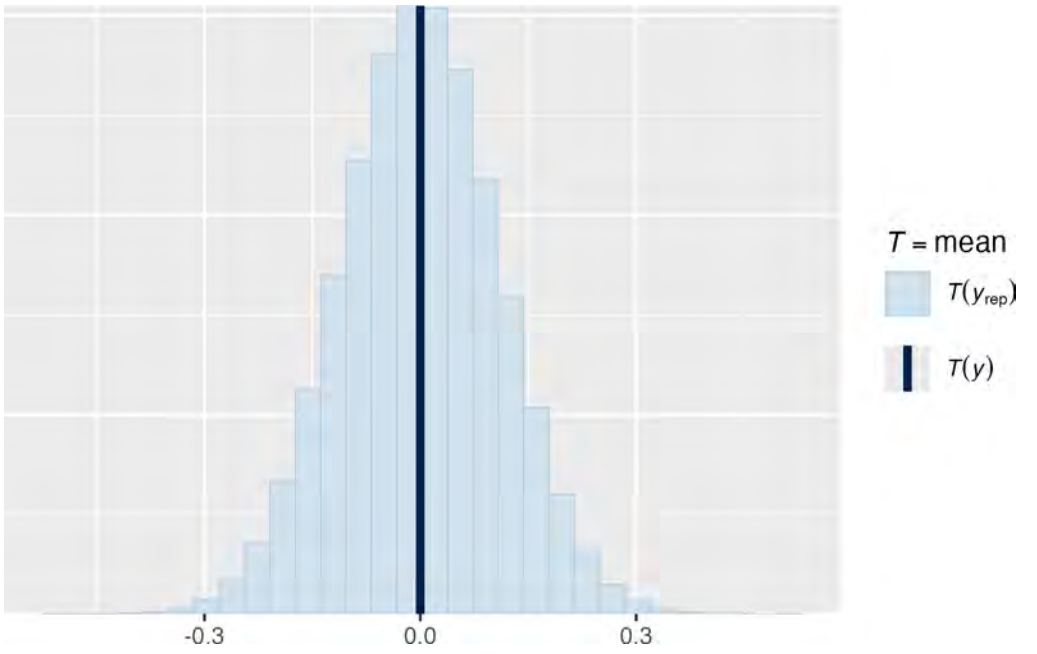

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.
```



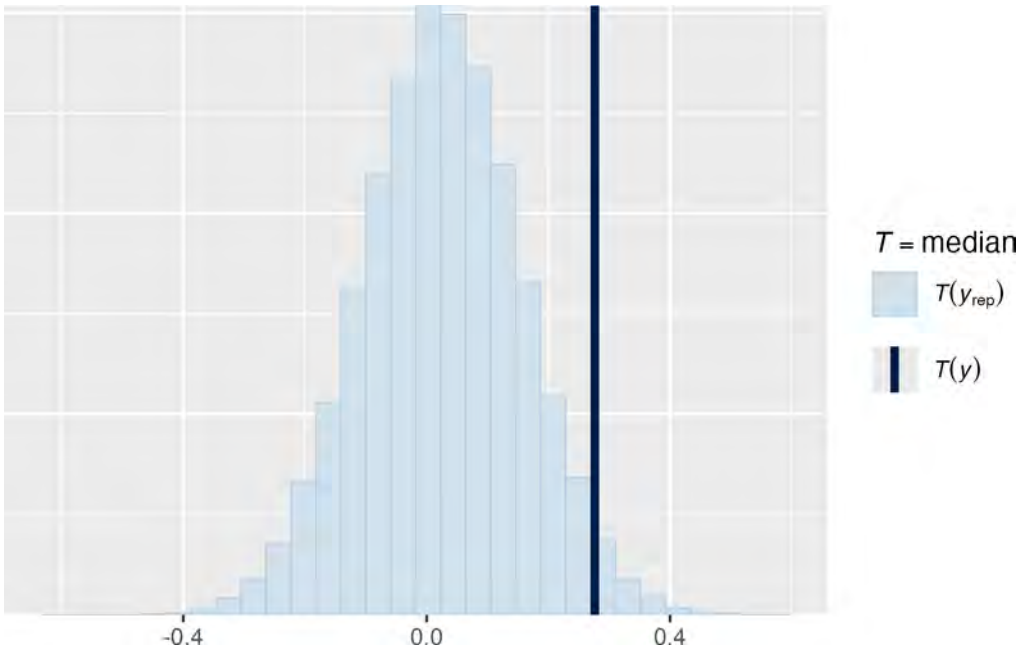

```
=== PPC: Mean ===
```

```
Observed mean (dark line) should fall within the distribution of replicated means.
```



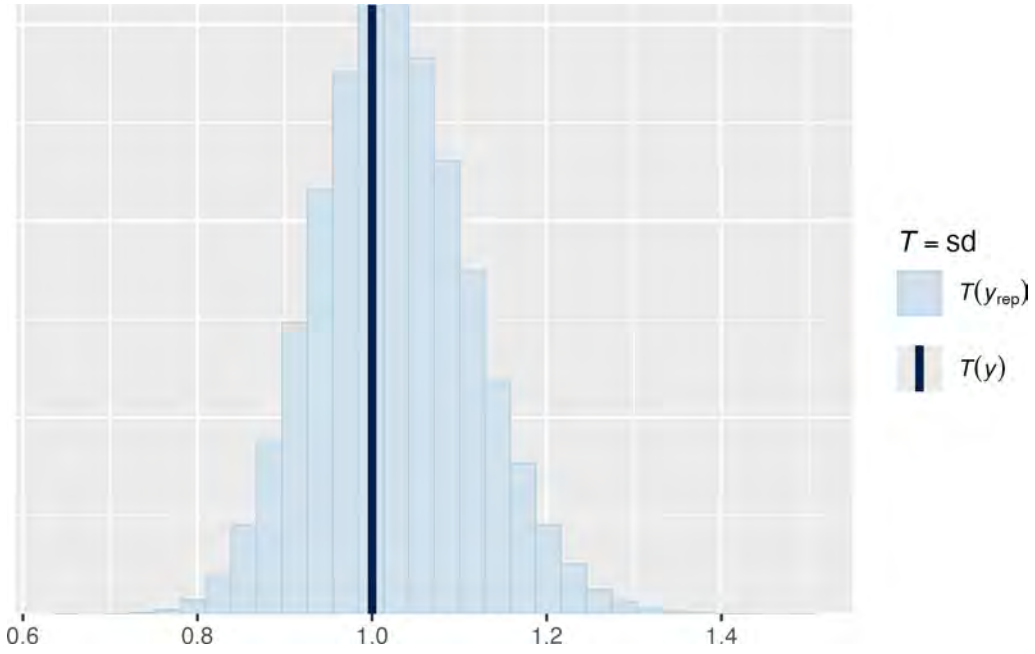
```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.
```



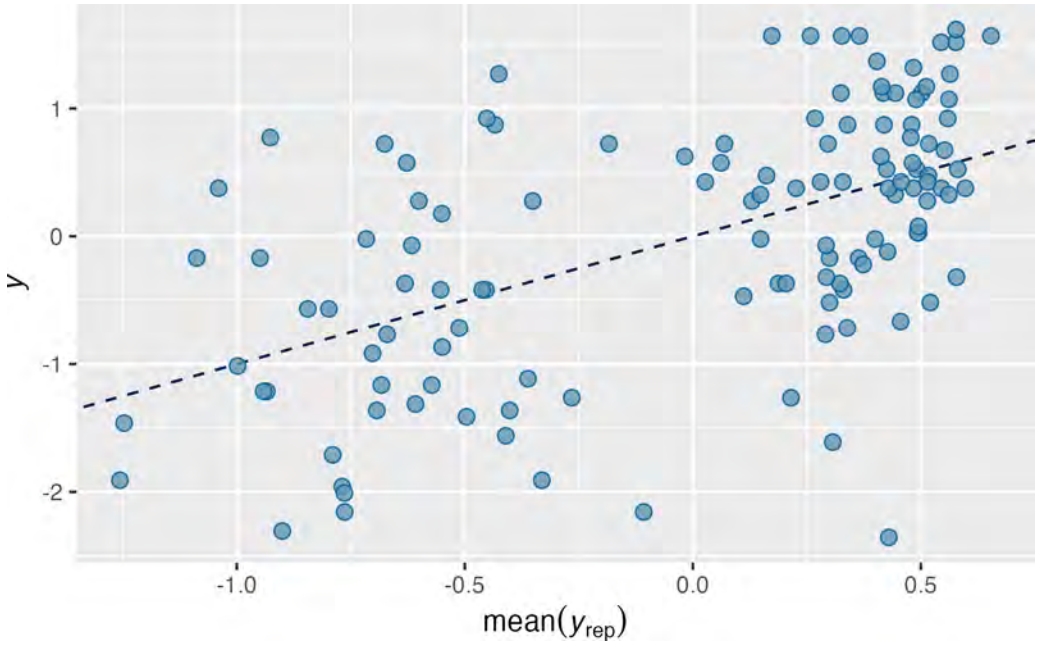

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.
```



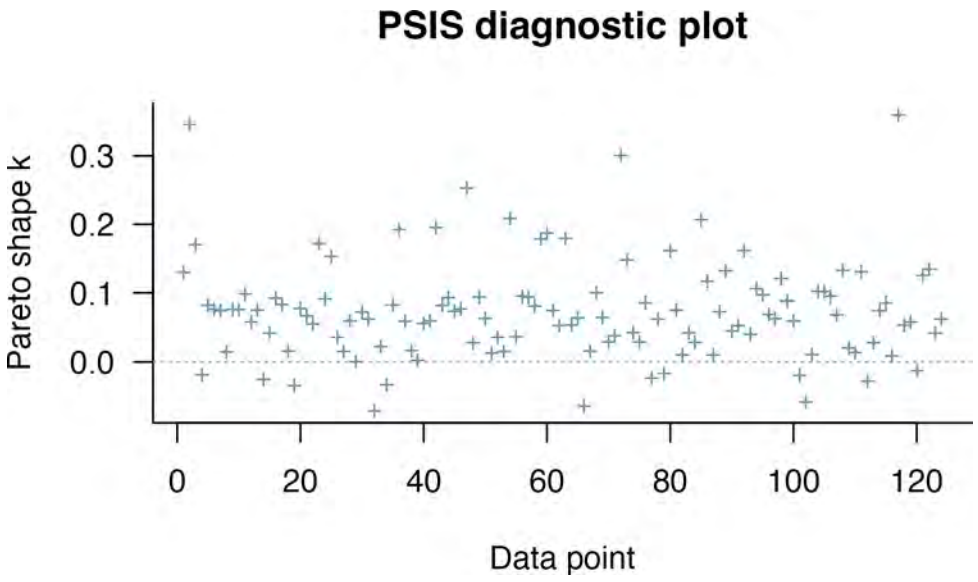

```
=== PPC: Scatter Average ===
```

```
Points should cluster around the diagonal. Systematic patterns indicate misfit.
```



```
=== Pareto-k ===

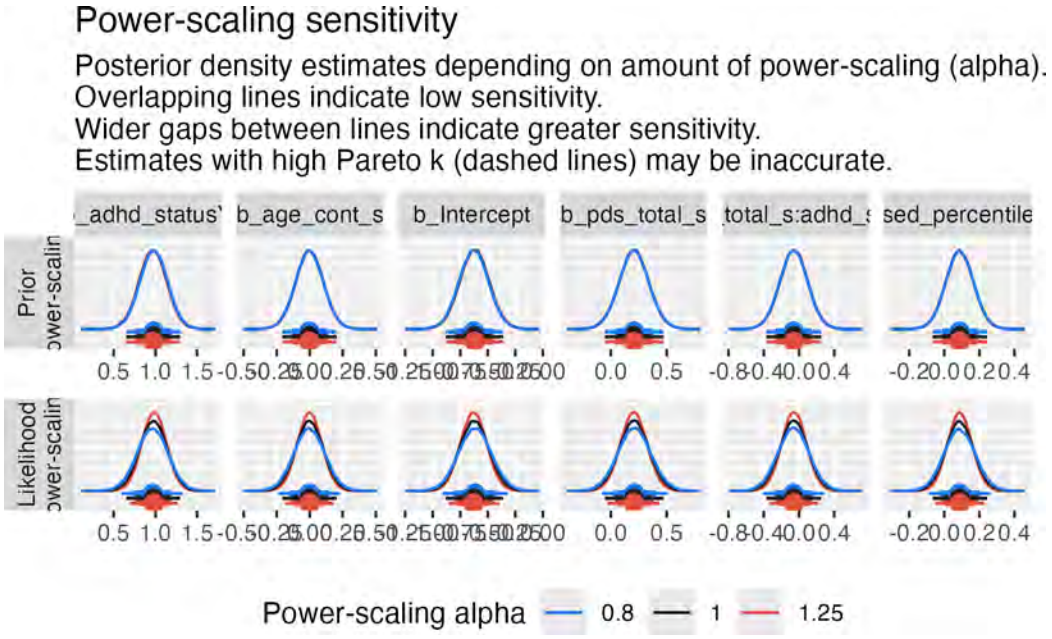

All Pareto k estimates are good (k < 0.7).
```

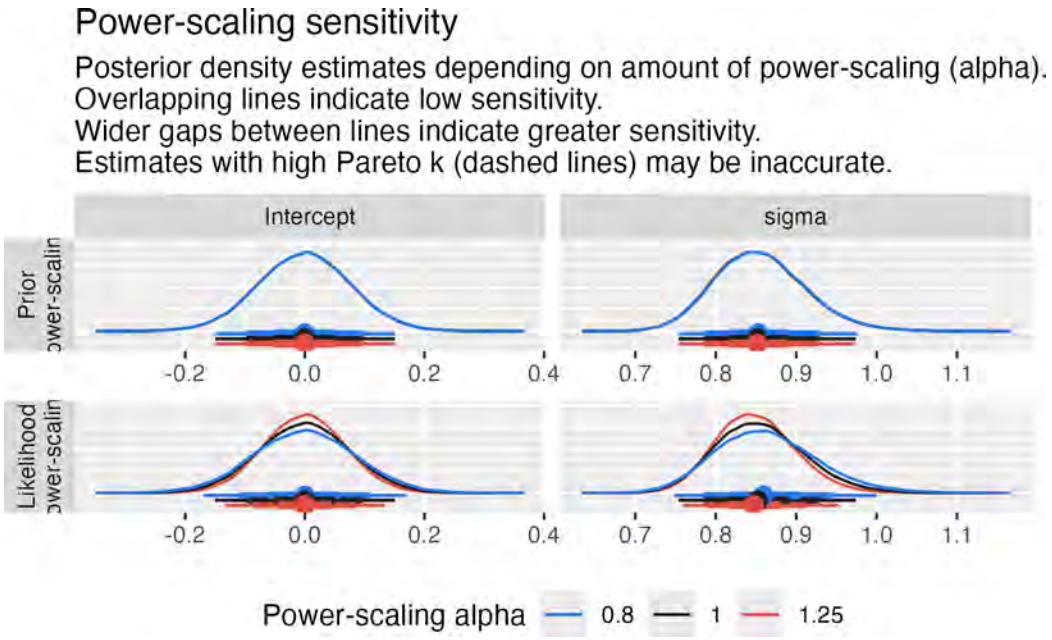



```
=== Power-Scaled Sensitivity Checks ===


Sensitivity based on cjs_dist
Prior selection: all priors
Likelihood selection: all data

                  variable prior likelihood diagnosis
               b_Intercept 0.018      0.098         -
             b_pds_total_s 0.003      0.086         -
            b_adhd_statusY 0.021      0.106         -
              b_age_cont_s 0.003      0.080         -
       b_sed_percentile_s 0.004      0.081         -
 b_pds_total_s:adhd_statusY 0.004      0.080         -
                     sigma 0.009      0.148         -
                 Intercept 0.001      0.079         -
```

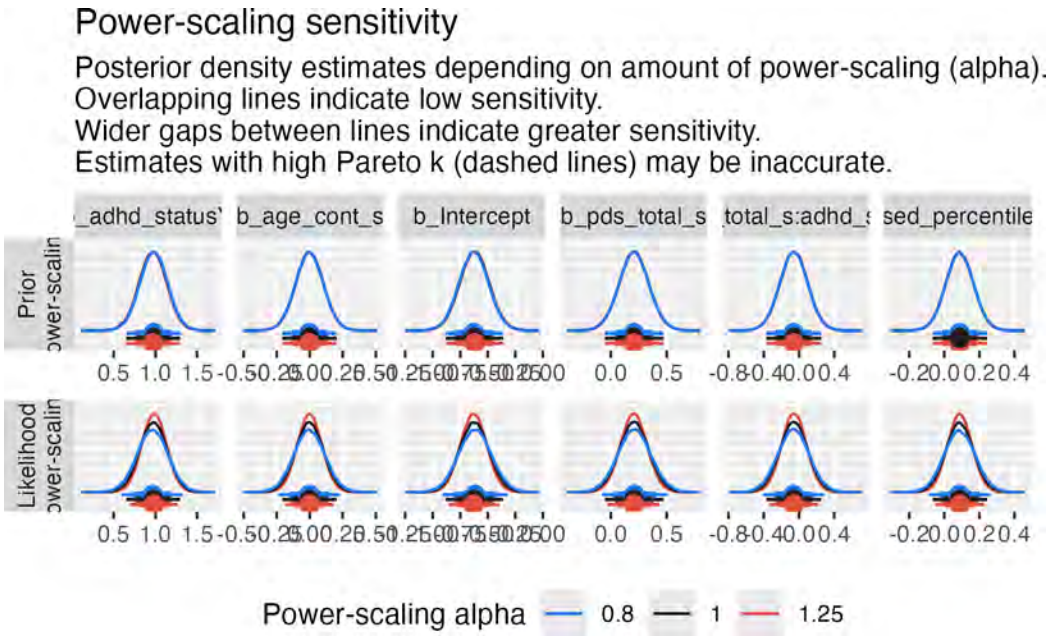


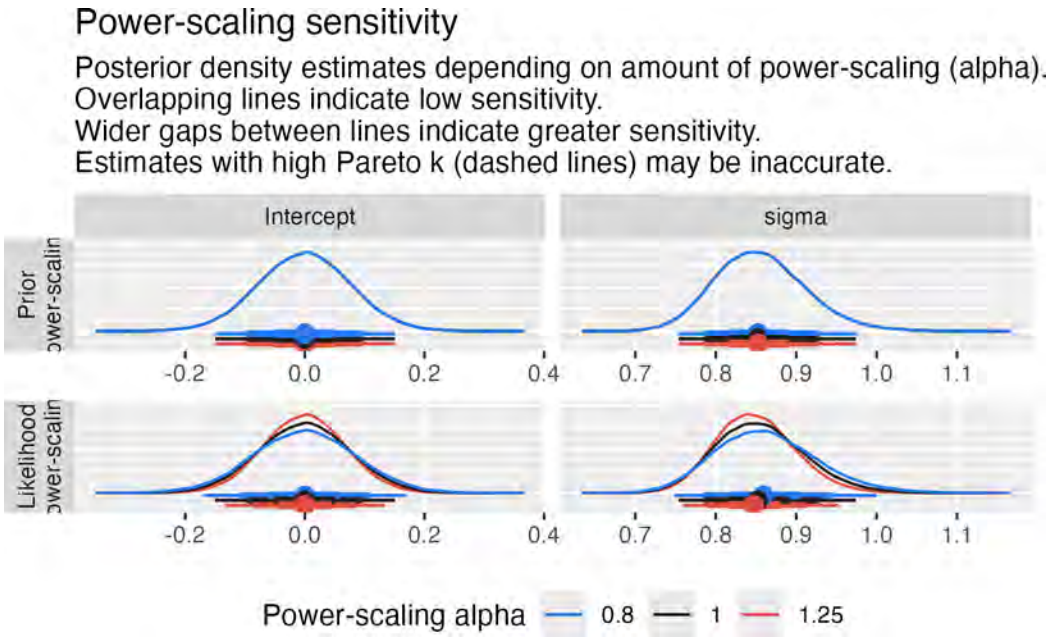



```
Sensitivity based on cjs_dist
Prior selection: b
Likelihood selection: all data

                variable prior likelihood diagnosis
           b_Intercept 0.018      0.098         -
```

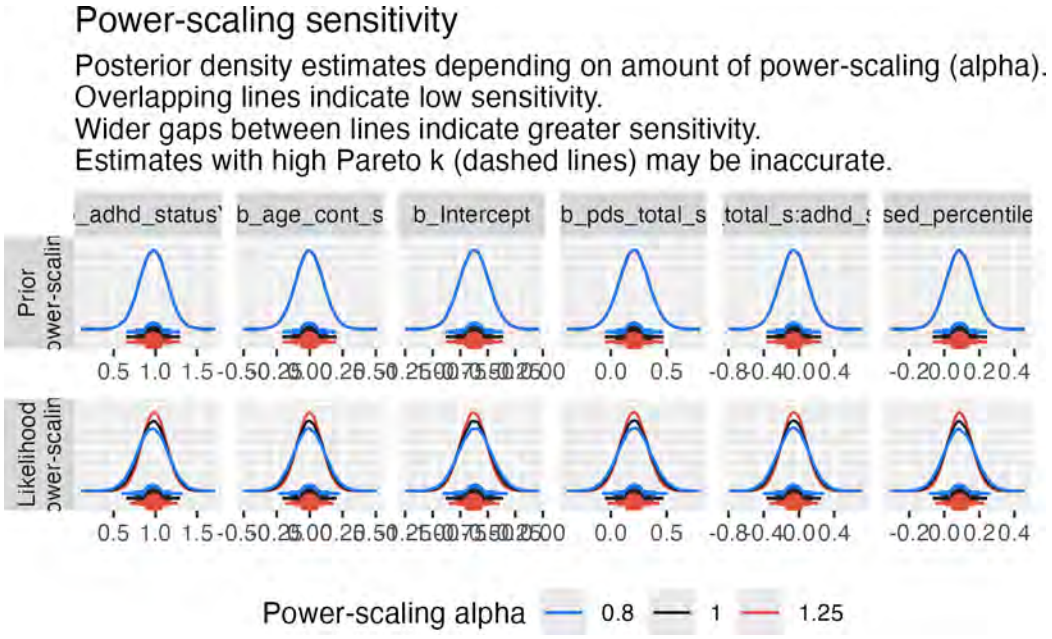
b_pds_total_s	0.003	0.086	-
b_adhd_statusY	0.021	0.106	-
b_age_cont_s	0.003	0.080	-
b_sed_percentile_s	0.003	0.081	-
b_pds_total_s:adhd_statusY	0.004	0.080	-
sigma	0.000	0.148	-
Intercept	0.000	0.079	-

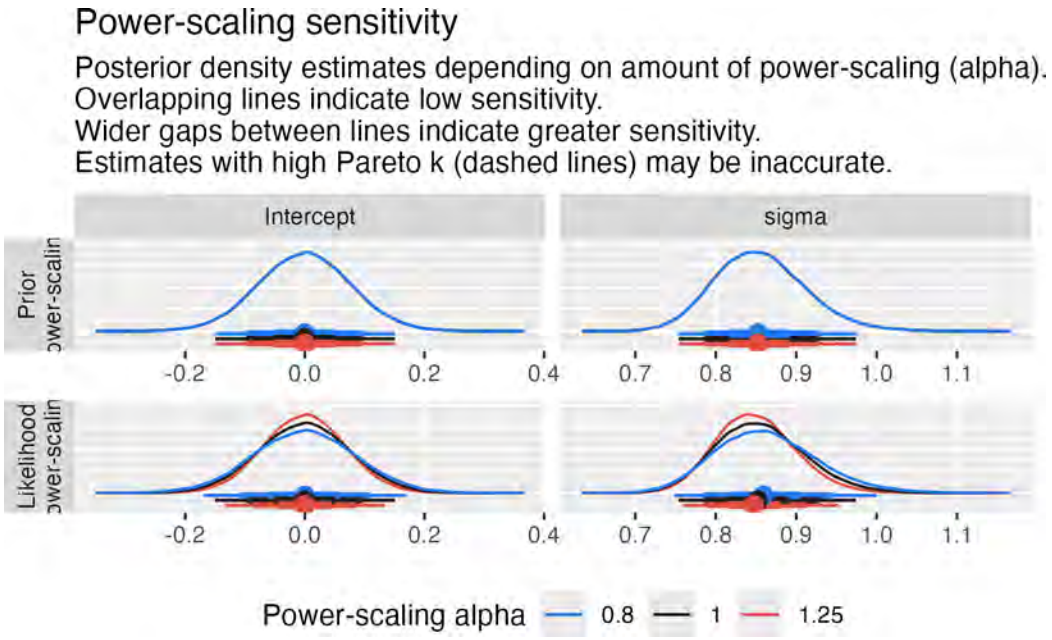





```
Sensitivity based on cjs_dist
Prior selection: intercept
Likelihood selection: all data
```

variable	prior	likelihood	diagnosis
b_Intercept	0	0.098	-
b_pds_total_s	0	0.086	-
b_adhd_statusY	0	0.106	-
b_age_cont_s	0	0.080	-
b_sed_percentile_s	0	0.081	-
b_pds_total_s:adhd_statusY	0	0.080	-
sigma	0	0.148	-
Intercept	0	0.079	-

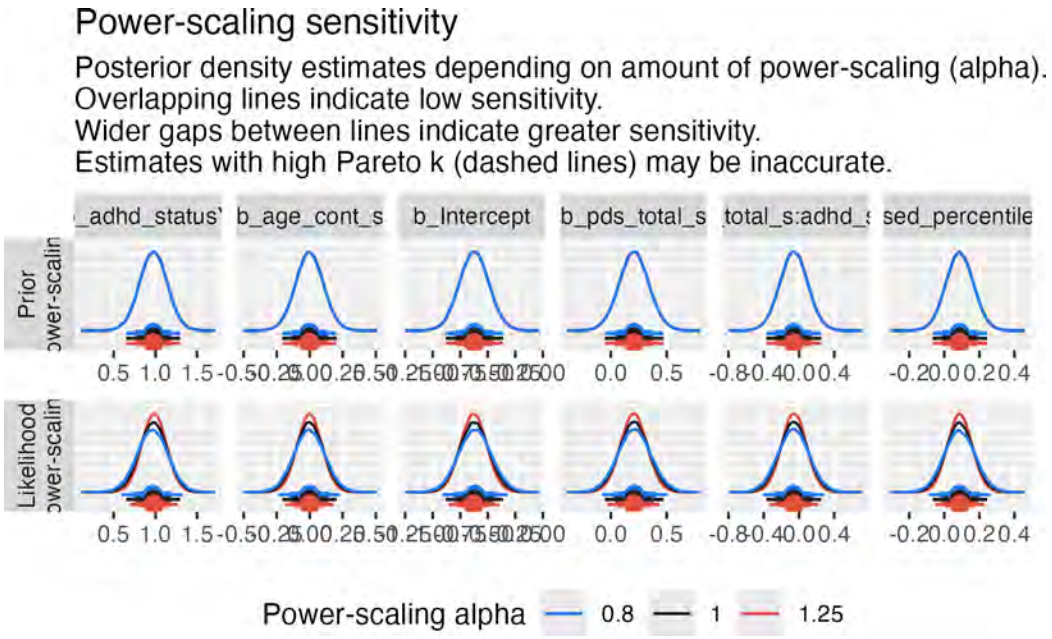


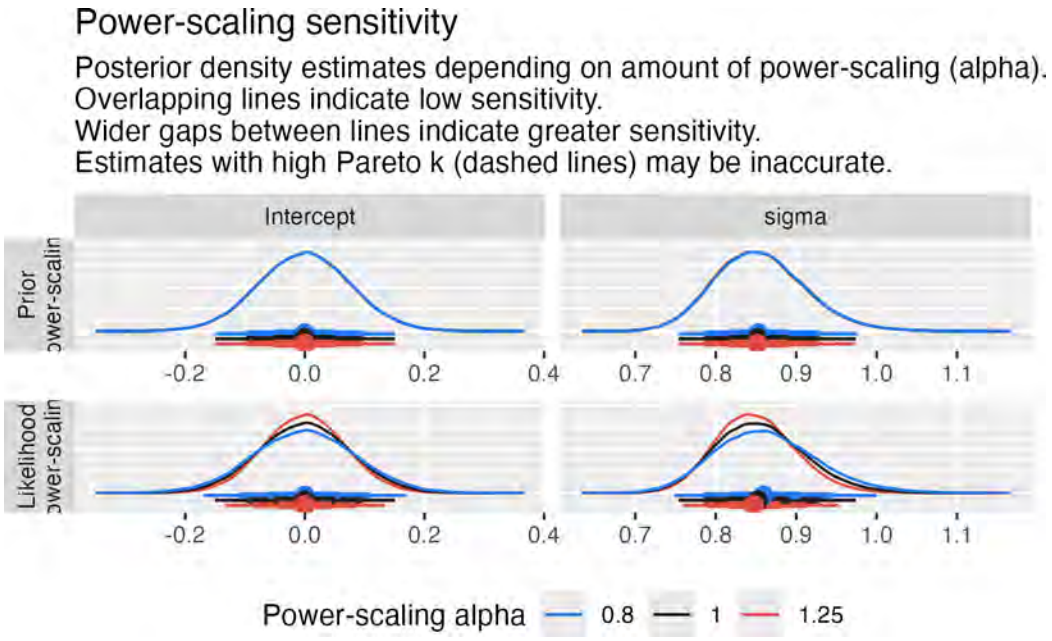



```
Sensitivity based on cjs_dist
Prior selection: sigma
Likelihood selection: all data

                variable prior likelihood diagnosis
           b_Intercept 0.001      0.098         -
```

b_pds_total_s	0.001	0.086	-
b_adhd_statusY	0.001	0.106	-
b_age_cont_s	0.001	0.080	-
b_sed_percentile_s	0.001	0.081	-
b_pds_total_s:adhd_statusY	0.001	0.080	-
sigma	0.009	0.148	-
Intercept	0.001	0.079	-





H1 Model no interaction output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3026468 0.05595102 0.1882161 0.4057285
```

```
Summary of Posterior Distribution

Parameter           |      Mean |    Median |             CI |      pd
-------------------------------------------------------------------------
b_Intercept         |     -0.63 |     -0.63 | [-0.89, -0.38] | 100.00%
b_pds_total_s       |      0.18 |      0.18 | [-0.03,  0.38] |  95.32%
b_adhd_statusY      |      0.98 |      0.98 | [ 0.66,  1.31] |    100%
b_age_cont_s        | -4.51e-03 | -4.68e-03 | [-0.21,  0.20] |  51.85%
b_sed_percentile_s  |      0.08 |      0.08 | [-0.07,  0.24] |  86.24%

Parameter           | % in ROPE | meaningful
-----------------------------------------
b_Intercept         | 7.50e-03% |       TRUE
b_pds_total_s       |    22.78% |      FALSE
b_adhd_statusY      |        0% |       TRUE
b_age_cont_s        |    66.56% |      FALSE
b_sed_percentile_s  |    57.15% |      FALSE
```

```
=== ADHD difference in raw outcome units ===
```

adhd_raw_effect_mean	adhd_raw_effect_lower	adhd_raw_effect_upper
19.74	13.36	26.27

H1 Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

# A tibble: 12 x 4
   variable            rhat ess_bulk ess_tail
   <chr>              <dbl>    <dbl>    <dbl>
 1 b_Intercept         1.00  160944.   91734.
 2 b_pds_total_s       1.00  107581.   94629.
 3 b_adhd_statusY      1.00  153956.   91829.
 4 b_age_cont_s        1.00   41694.   94265.
 5 b_sed_percentile_s  1.00   25796.   90641.
 6 sigma               1.00  152427.   90916.
 7 Intercept           1.00  171505.   89323.
 8 lprior              1.00  159130.   90694.
 9 lprior_b            1.00  155826.   91590.
10 lprior_intercept    1.00   58787.   63941.
11 lprior_sigma        1.00  152427.   90916.
12 lp__                1.04    1935.    2278.


=== R-hat & ESS for log posterior density ===

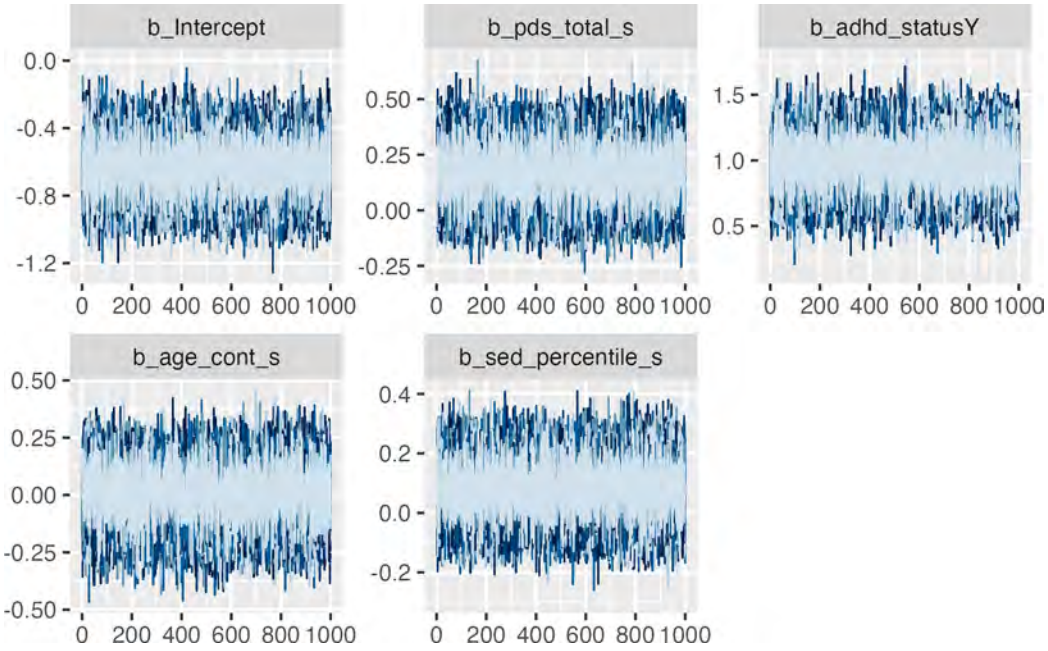
      rhat          ess_bulk       ess_tail
 Min.   :1.000   Min.   :1581   Min.   :2233
 1st Qu.:1.001   1st Qu.:1772   1st Qu.:2421
 Median :1.001   Median :1803   Median :2492
 Mean   :1.001   Mean   :1821   Mean   :2530
 3rd Qu.:1.002   3rd Qu.:1892   3rd Qu.:2657
 Max.   :1.004   Max.   :2046   Max.   :2813


=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

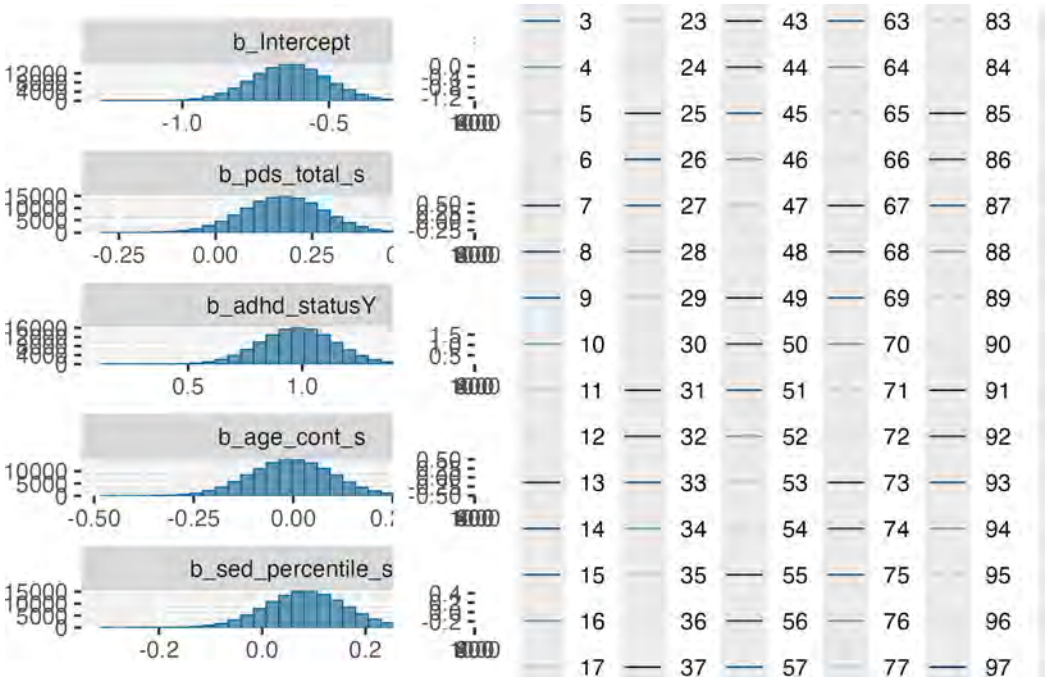
Priors specified for the model.

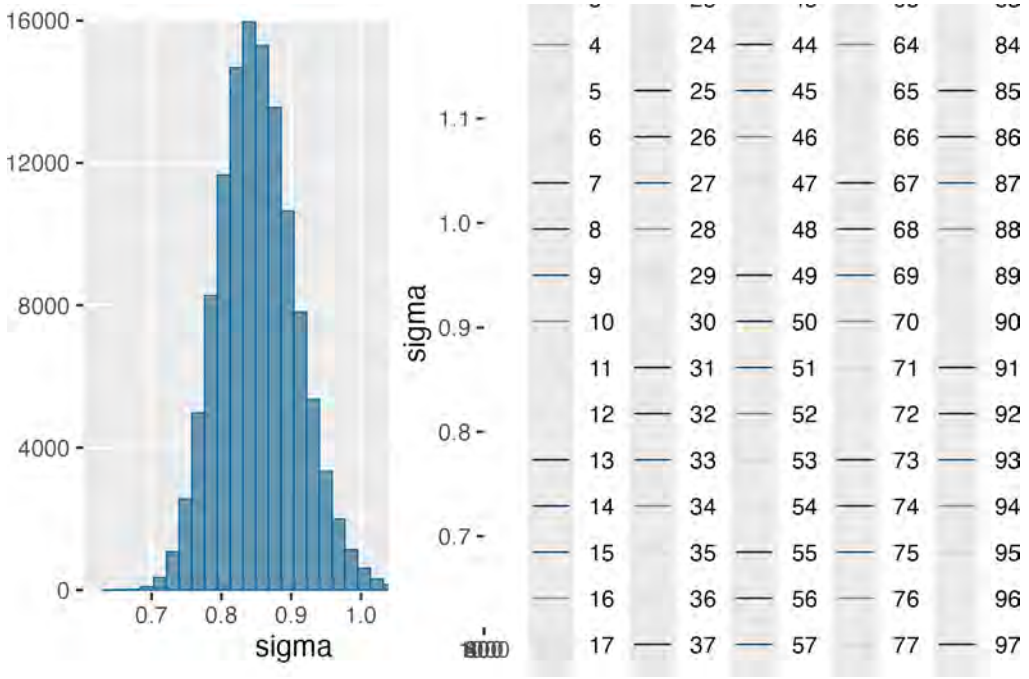
          prior     class             coef group resp dpar nlpar lb ub
   normal(0, 1)         b
   normal(0, 1)         b     adhd_statusY
   normal(0, 1)         b       age_cont_s
   normal(0, 1)         b      pds_total_s
   normal(0, 1)         b sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                          0
       tag       source
         b         user
```

```
         b (vectorized)
         b (vectorized)
         b (vectorized)
         b (vectorized)
 intercept         user
     sigma         user
```

```
=== Posterior Predictive Checks ===
```

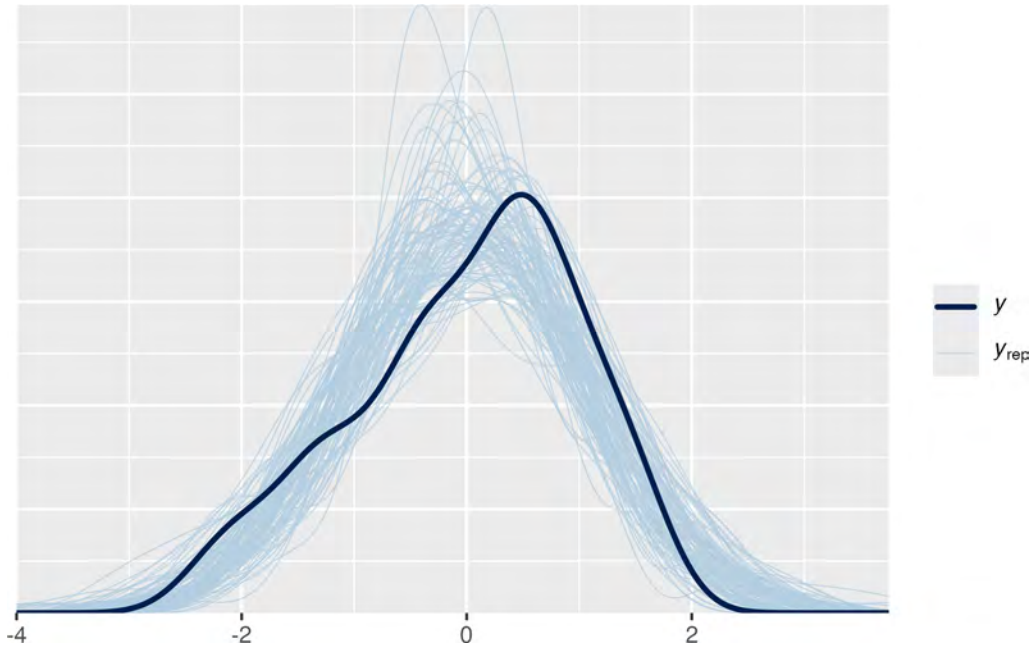
Density plots for each parameter.





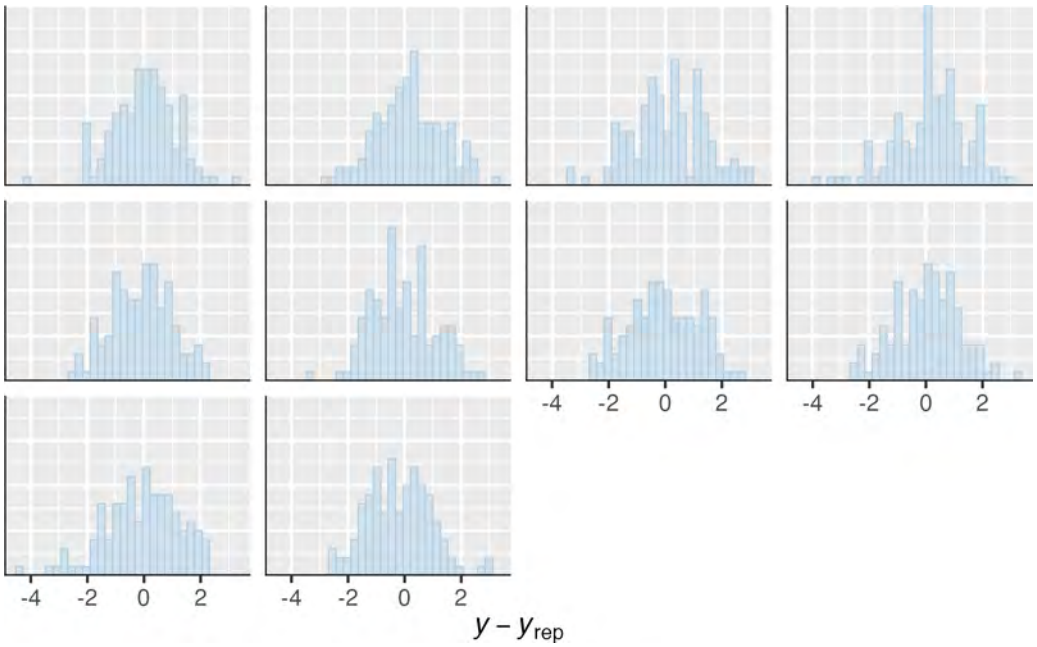

```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.
```



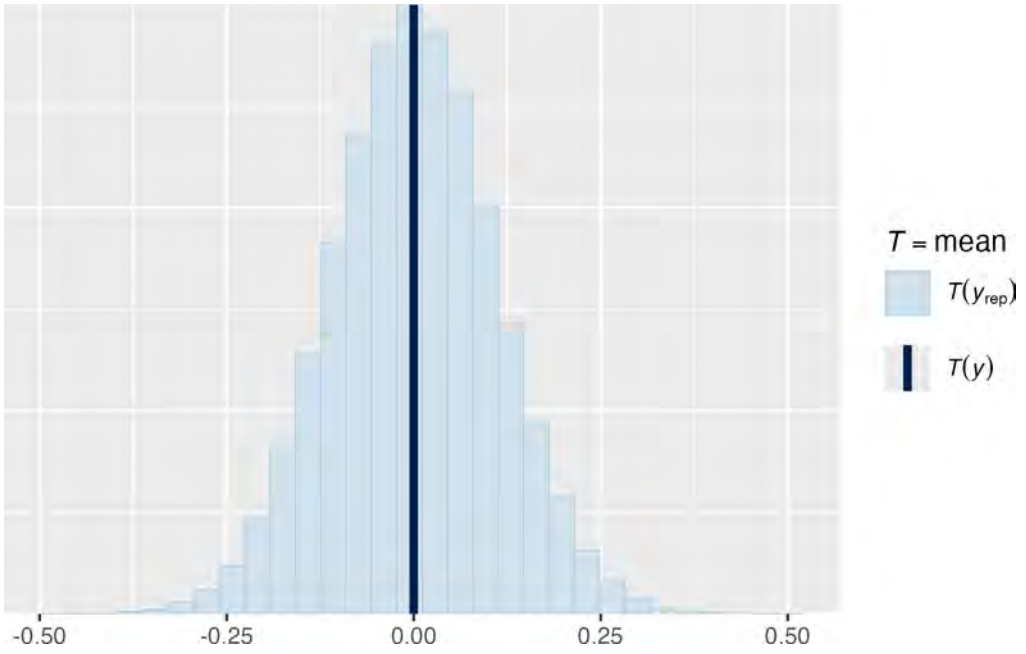
```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.
```



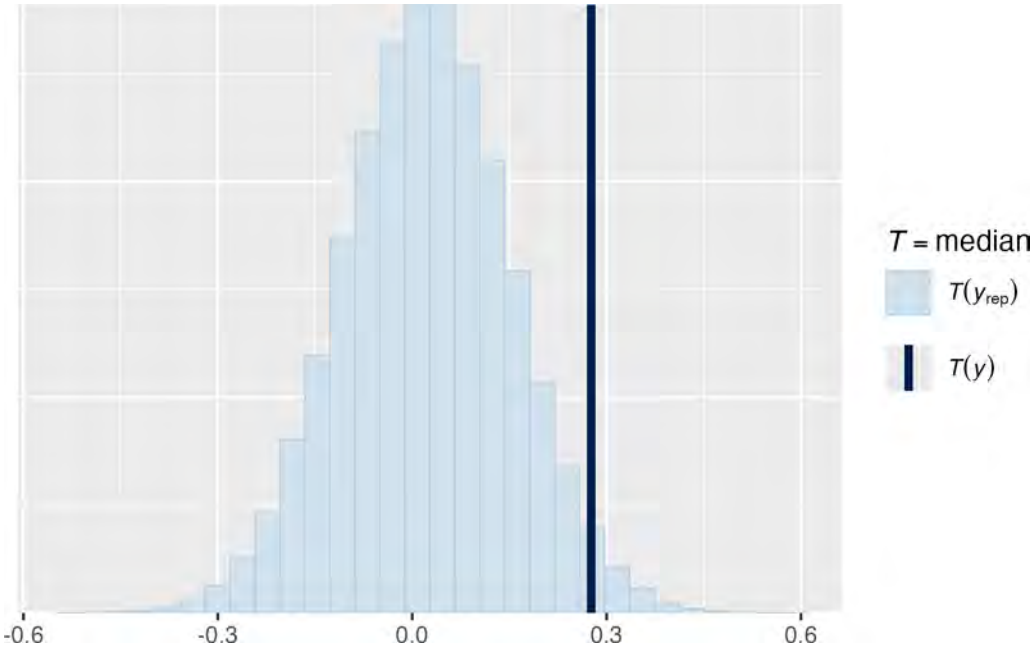

```
=== PPC: Mean ===

Observed mean (dark line) should fall within the distribution of replicated means.
```



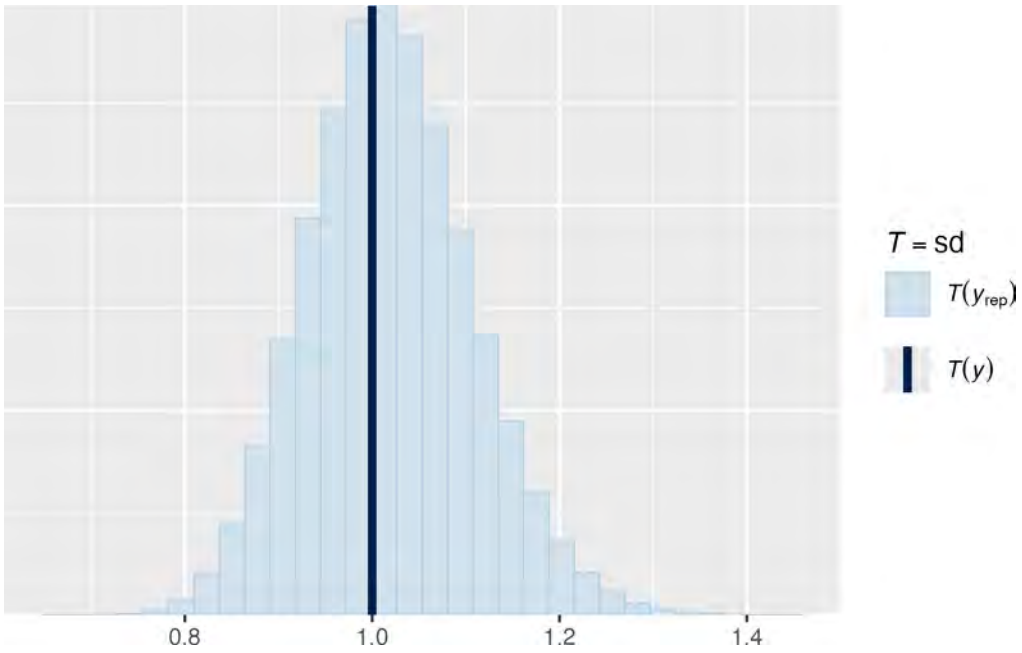

```
=== PPC: Median ===
```

```
Observed median (dark line) should fall within the distribution of replicated means.
```



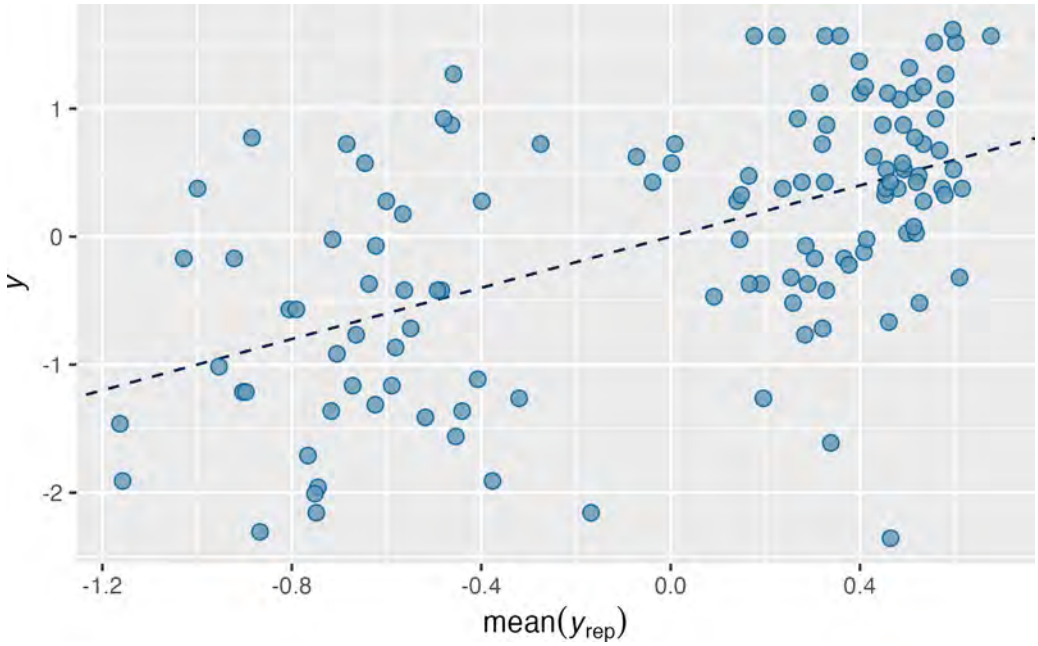
```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.
```



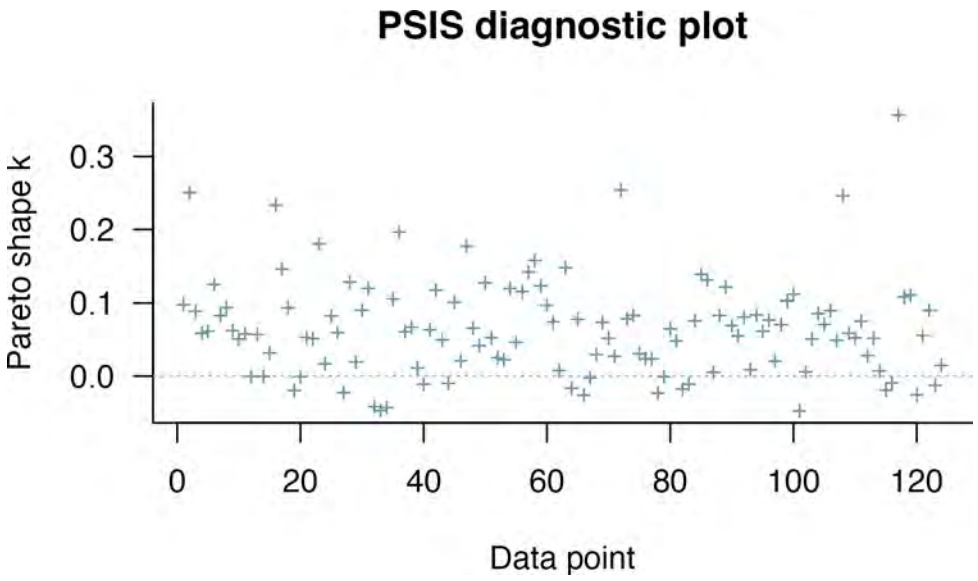

```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.
```




```
=== Pareto-k ===
```

```
All Pareto k estimates are good (k < 0.7).
```



H2 Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3167366 0.05578322 0.2012883 0.4181871
```

```
Summary of Posterior Distribution

Parameter                          |  Mean | Median |             CI |      pd
-------------------------------------------------------------------------------
b_Intercept                        | -0.62 |  -0.62 | [-0.88, -0.37] | 100.00%
b_pubertal_timing_s                |  0.31 |   0.31 | [ 0.09,  0.54] |  99.65%
b_adhd_statusY                     |  1.00 |   1.01 | [ 0.69,  1.32] |    100%
b_sed_percentile_s                 |  0.12 |   0.12 | [-0.04,  0.27] |  93.42%
b_pubertal_timing_s:adhd_statusY   | -0.34 |  -0.34 | [-0.65, -0.04] |  98.61%

Parameter                          | % in ROPE | meaningful
-----------------------------------------------------------
b_Intercept                        | 5.00e-03% |       TRUE
b_pubertal_timing_s                |     3.03% |       TRUE
b_adhd_statusY                     |        0% |       TRUE
b_sed_percentile_s                 |    41.02% |      FALSE
b_pubertal_timing_s:adhd_statusY   |     5.48% |      FALSE
```

H2 Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

# A tibble: 12 x 4
   variable                          rhat ess_bulk ess_tail
   <chr>                            <dbl>    <dbl>    <dbl>
 1 b_Intercept                       1.00  141595.   90602.
 2 b_pubertal_timing_s               1.00   83147.   89343.
 3 b_adhd_statusY                    1.00  141815.   90993.
 4 b_sed_percentile_s                1.00   23282.   90788.
 5 b_pubertal_timing_s:adhd_statusY  1.00   84665.   90134.
 6 sigma                             1.00  137352.   93133.
 7 Intercept                         1.00  153778.   90807.
 8 lprior                            1.00  142111.   90627.
 9 lprior_b                          1.00  128684.   90753.
```

```
10 lprior_intercept                  1.00   60996.   68970.
11 lprior_sigma                      1.00  137352.   93133.
12 lp__                              1.05    1459.    1798.


=== R-hat & ESS for log posterior density ===

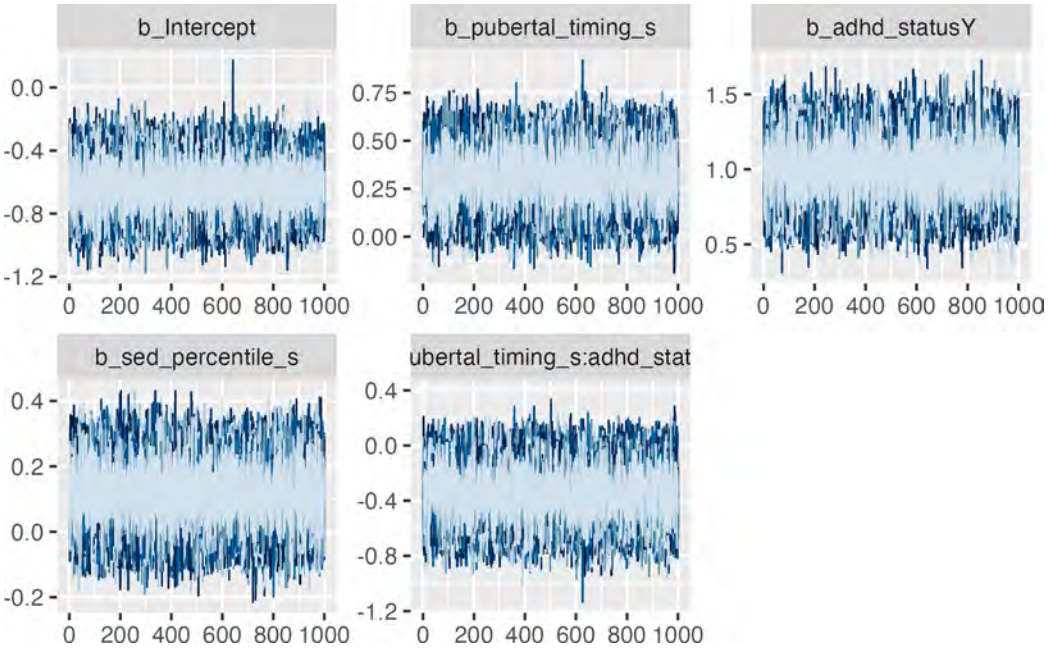
      rhat           ess_bulk       ess_tail
 Min.   :0.9998   Min.   :1578   Min.   :2350
 1st Qu.:1.0005   1st Qu.:1748   1st Qu.:2501
 Median :1.0010   Median :1829   Median :2618
 Mean   :1.0012   Mean   :1831   Mean   :2634
 3rd Qu.:1.0017   3rd Qu.:1923   3rd Qu.:2773
 Max.   :1.0036   Max.   :2013   Max.   :3026


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



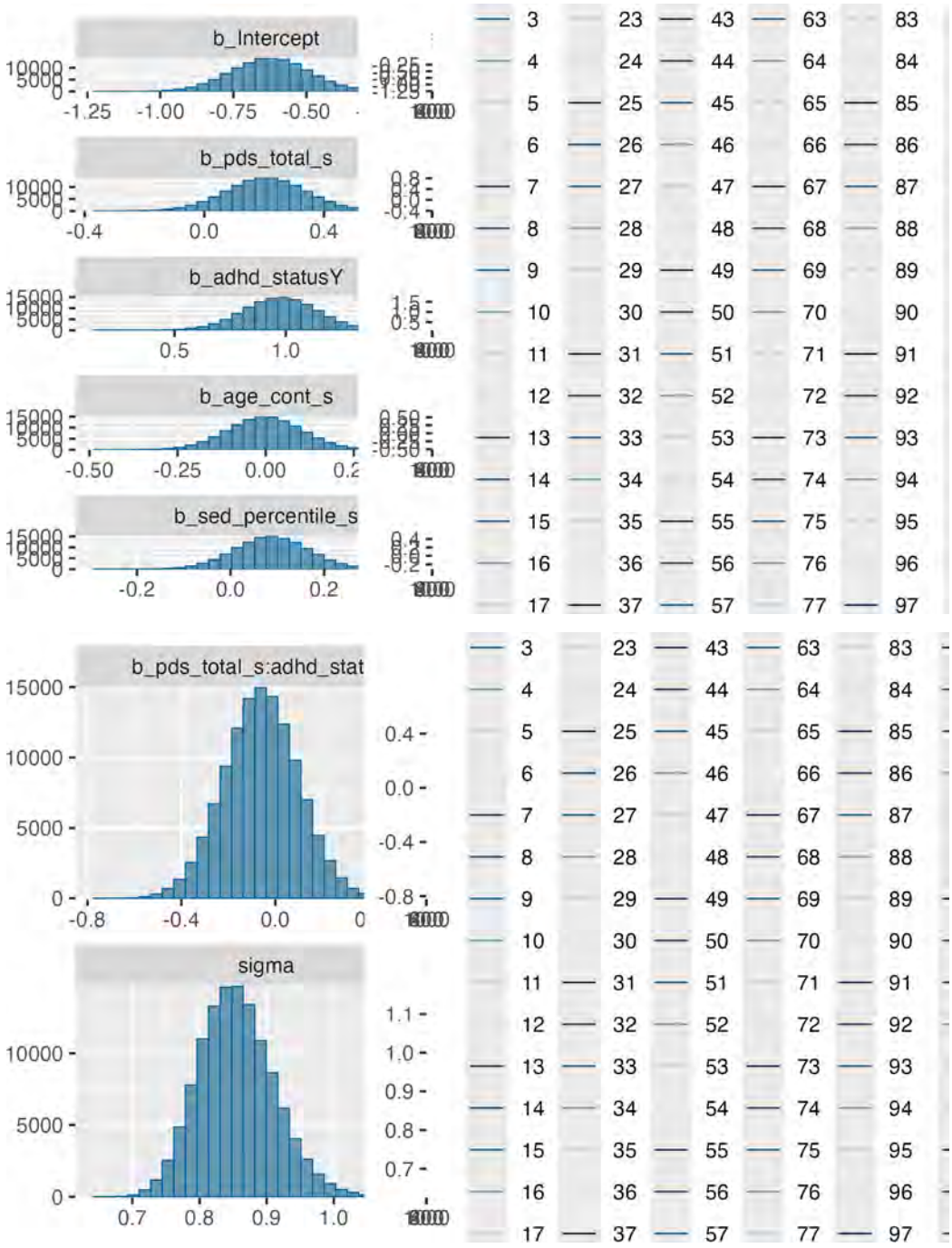
```
=== Prior Summary ===

Priors specified for the model.

          prior     class                           coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b                     adhd_statusY
   normal(0, 1)         b               pubertal_timing_s
   normal(0, 1)         b pubertal_timing_s:adhd_statusY
   normal(0, 1)         b                sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub       tag       source
              b         user
              b (vectorized)
              b (vectorized)
              b (vectorized)
              b (vectorized)
      intercept         user
  0       sigma         user

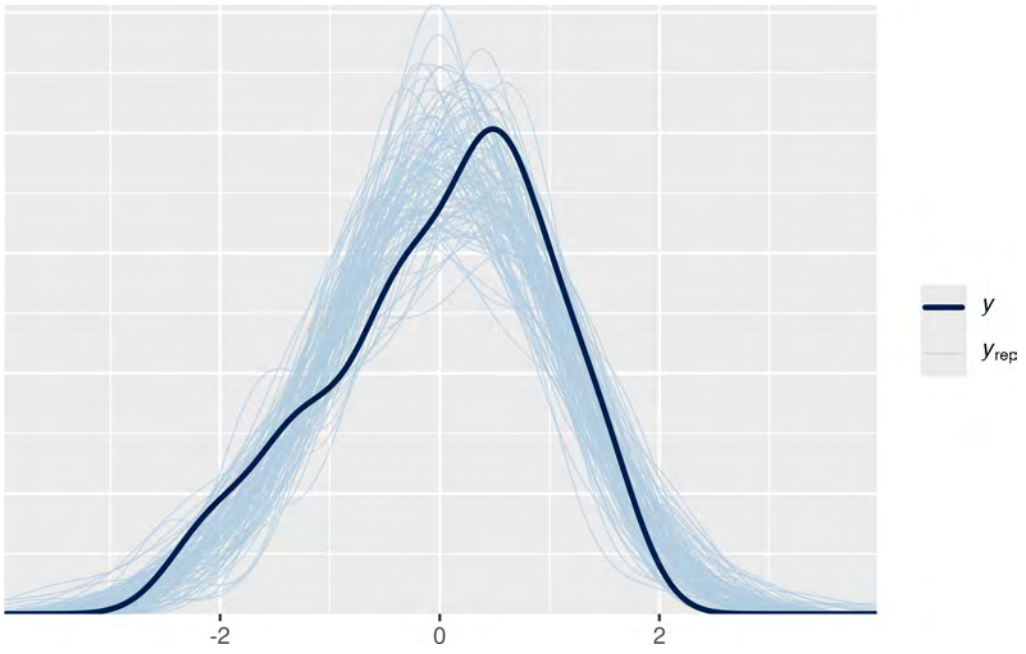

=== Posterior Predictive Checks ===

Density plots for each parameter.
```



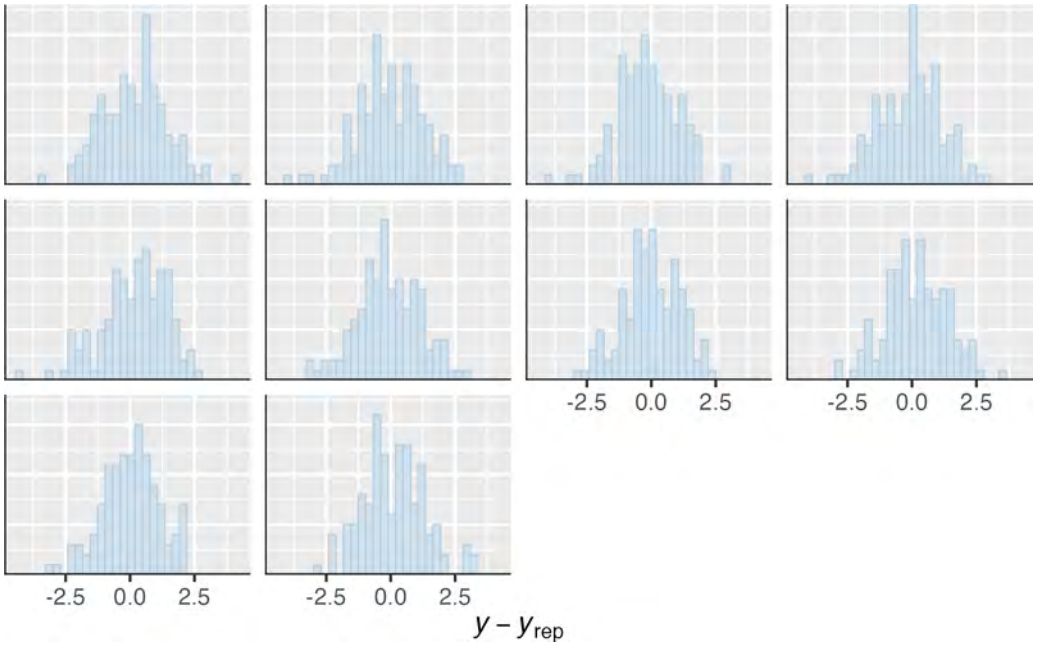
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.



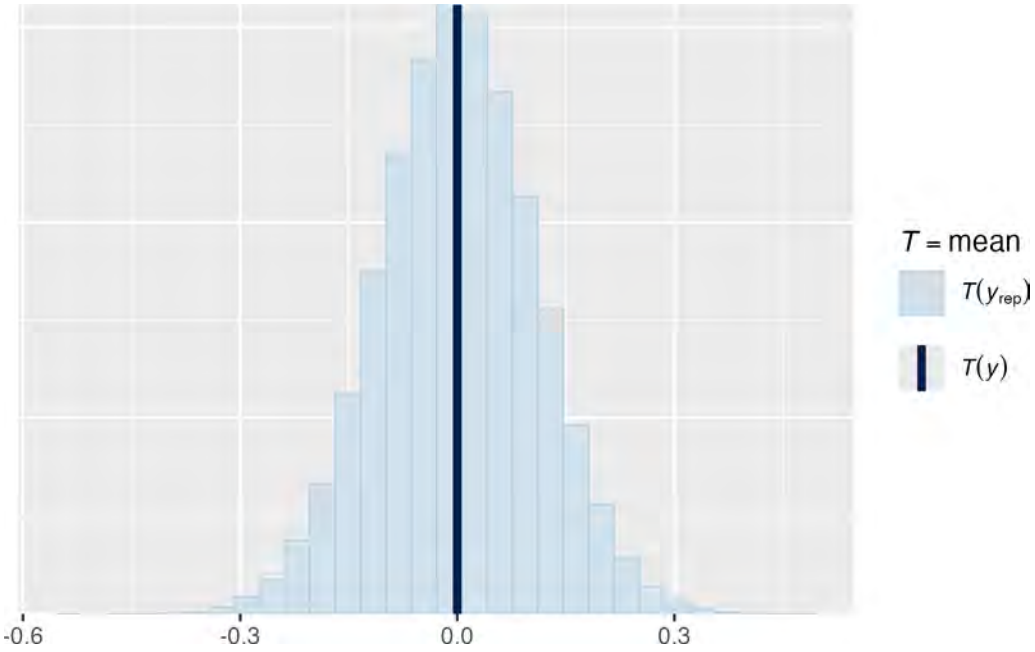

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.
```



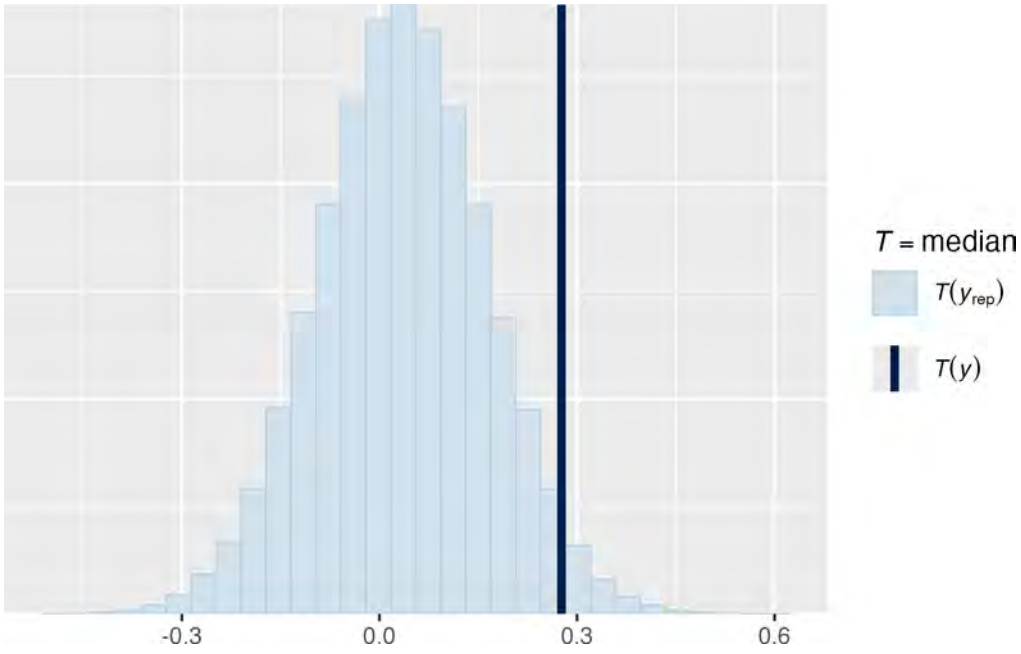
```
=== PPC: Mean ===
```

Observed mean (dark line) should fall within the distribution of replicated means.



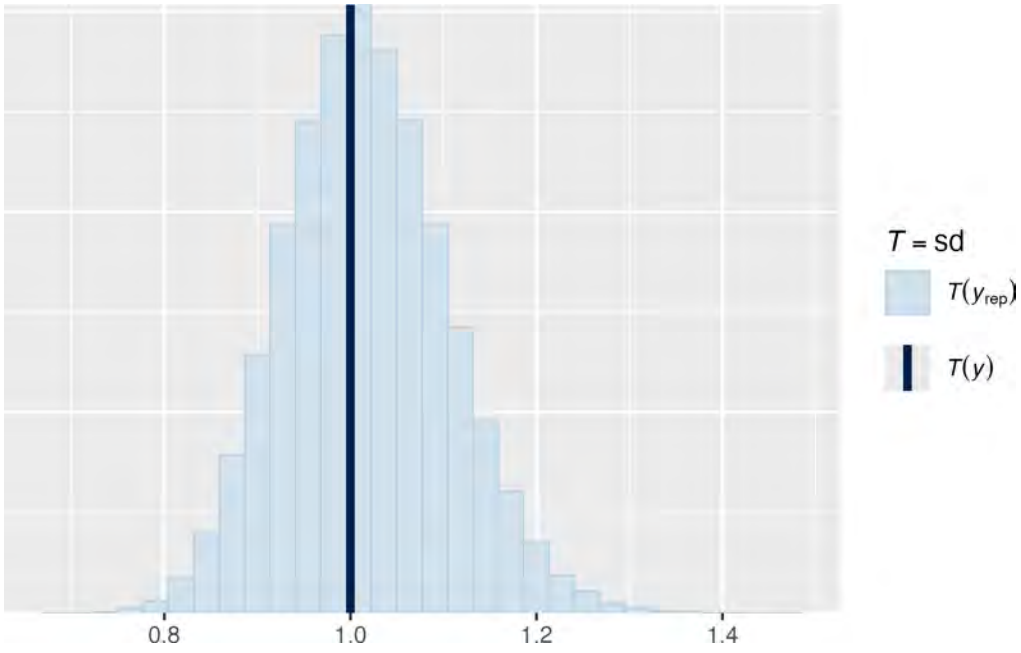

```
=== PPC: Median ===
```

Observed median (dark line) should fall within the distribution of replicated medians.



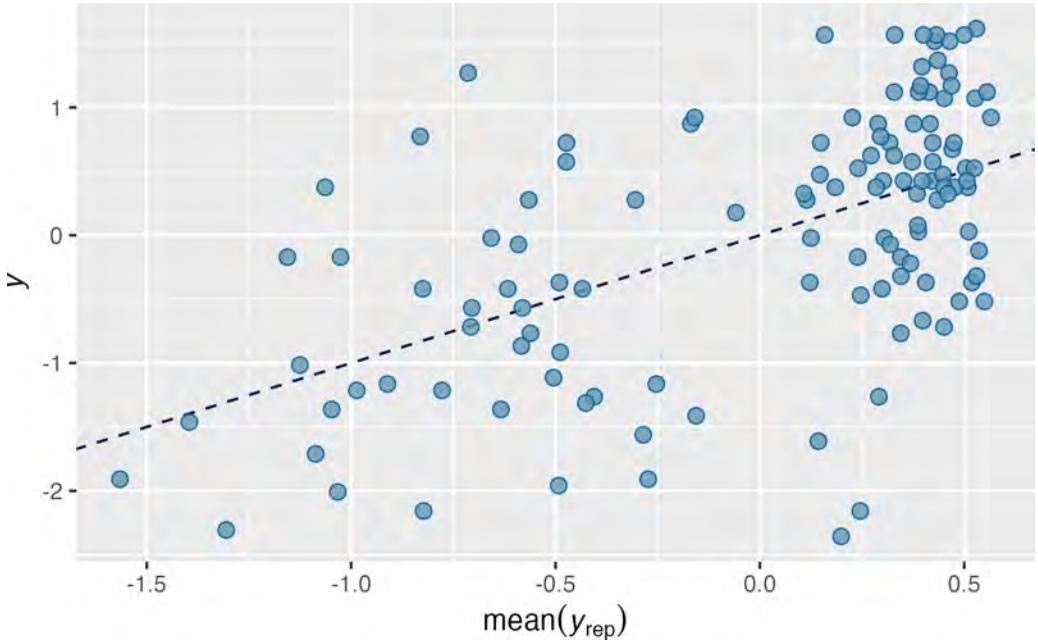

```
=== PPC: SD ===
```

```
Observed SD (dark line) should fall within the distribution of replicated SDs.
```



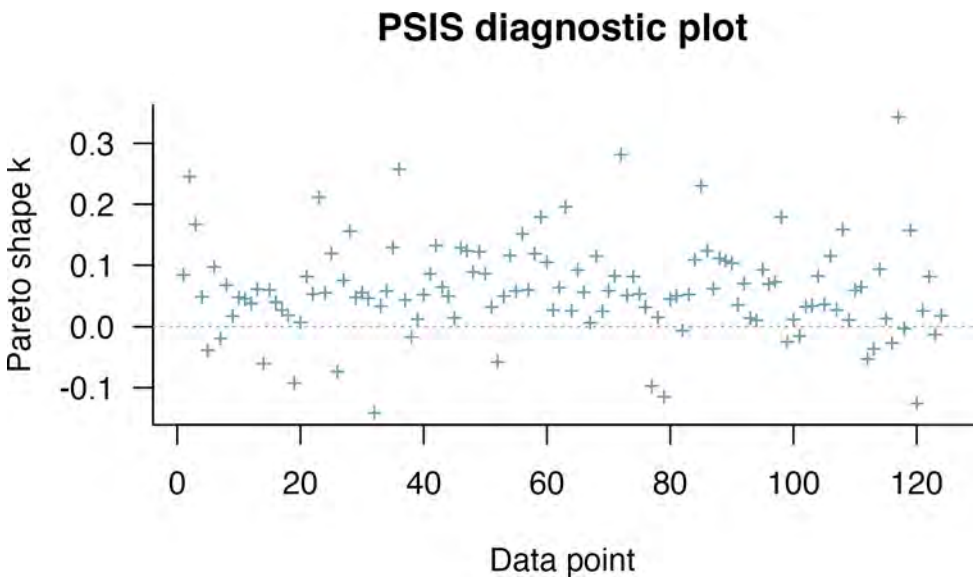
```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.
```




```
=== Pareto-k ===

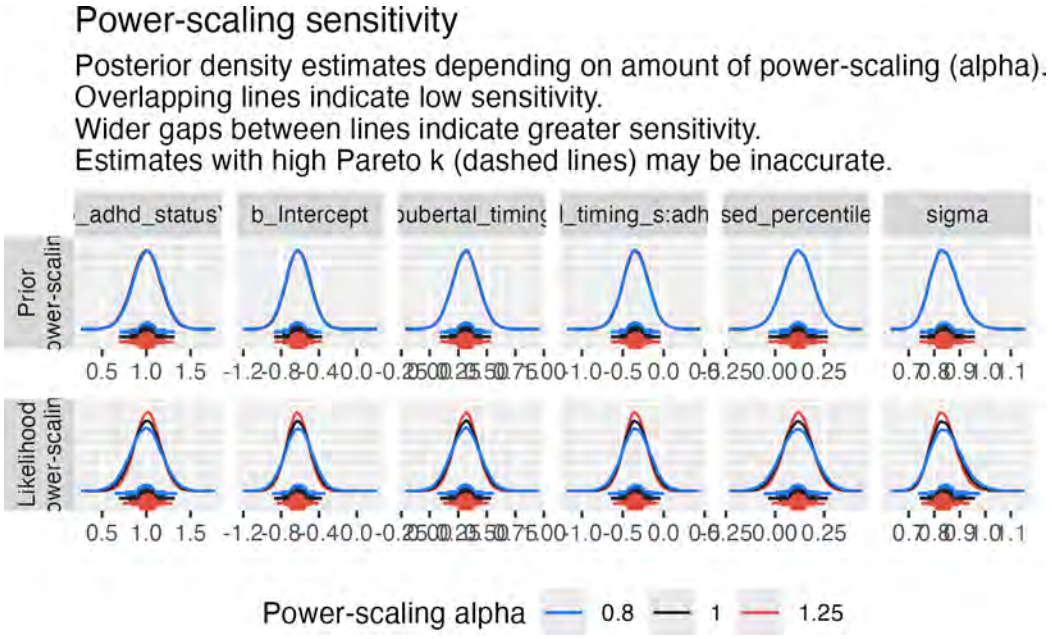
All Pareto k estimates are good (k < 0.7).
```

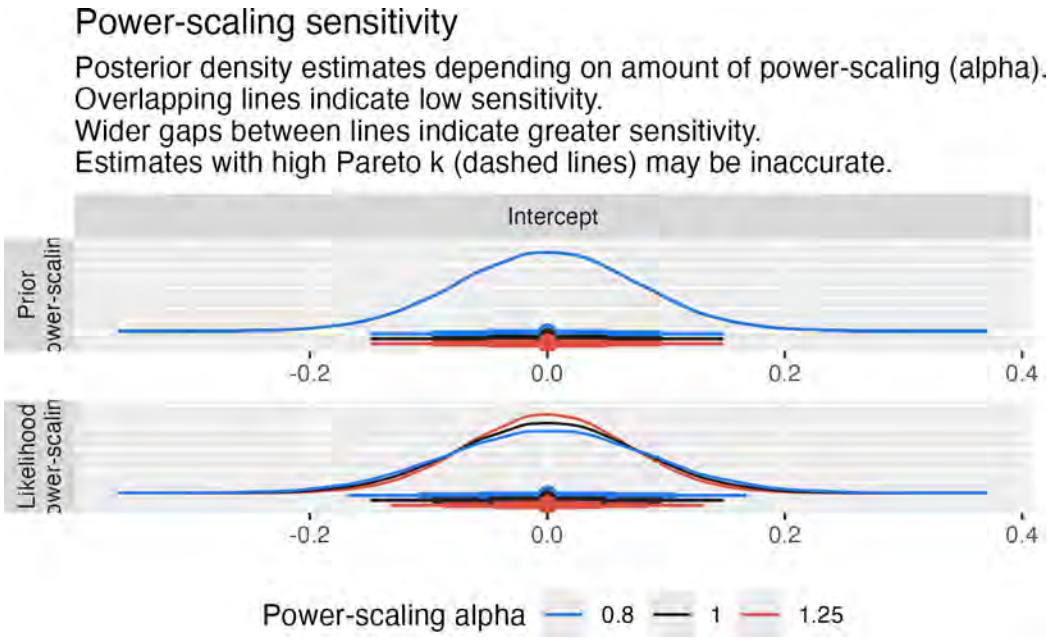



```
=== Power-Scaled Sensitivity Checks ===

Sensitivity based on cjs_dist
Prior selection: all priors
Likelihood selection: all data

                         variable prior likelihood diagnosis
                      b_Intercept 0.014      0.096         -
             b_pubertal_timing_s 0.008      0.072         -
                  b_adhd_statusY 0.022      0.098         -
              b_sed_percentile_s 0.005      0.092         -
 b_pubertal_timing_s:adhd_statusY 0.011      0.080         -
                            sigma 0.009      0.139         -
                        Intercept 0.001      0.079         -
```

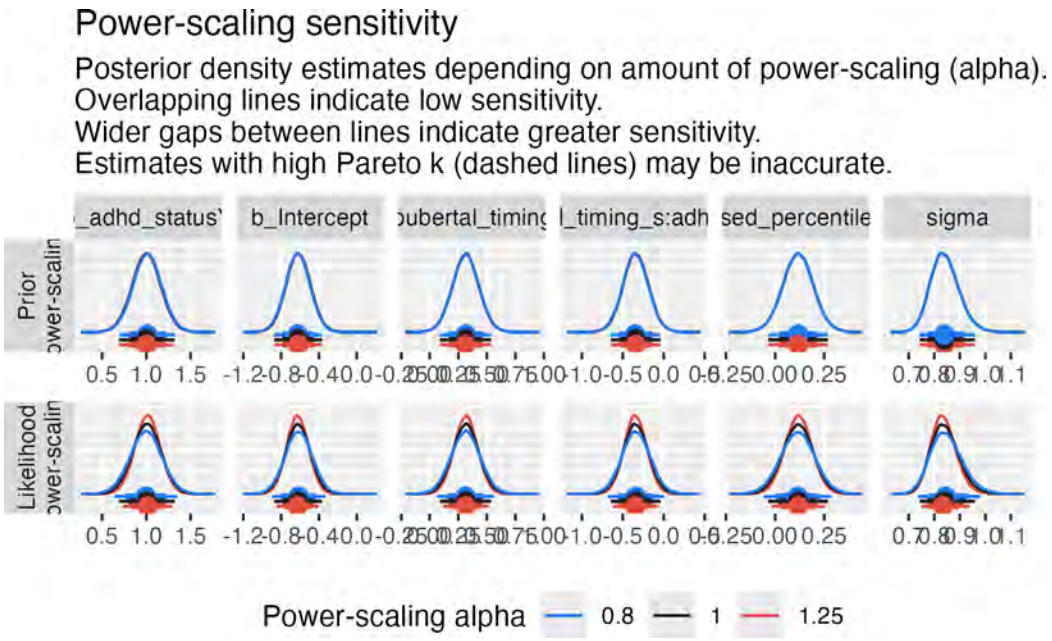


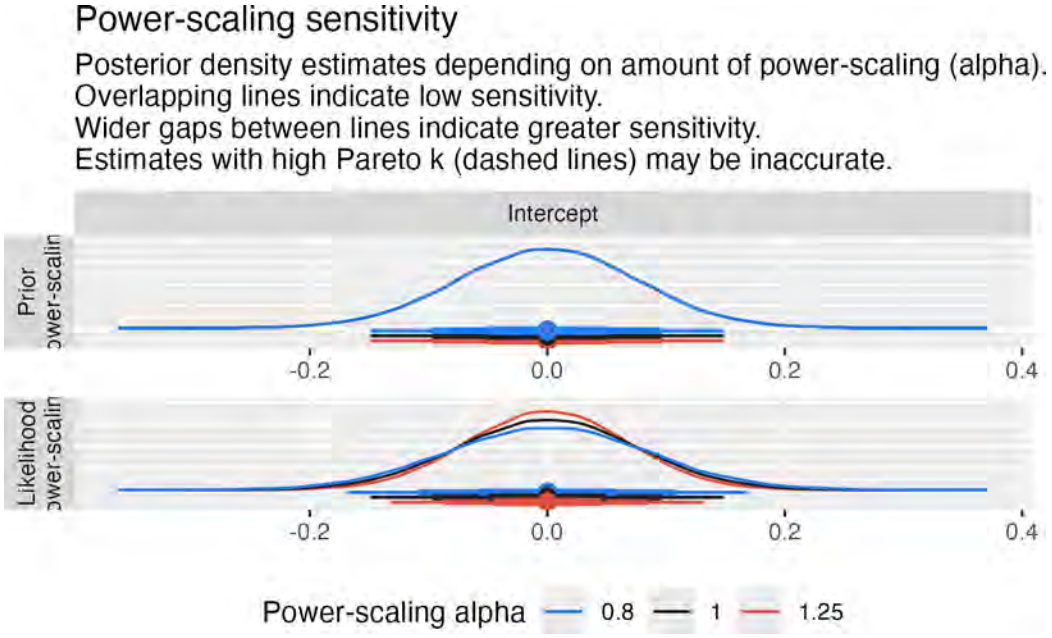



```
Sensitivity based on cjs_dist
Prior selection: b
Likelihood selection: all data

                    variable prior likelihood diagnosis
               b_Intercept 0.015      0.096         -
```

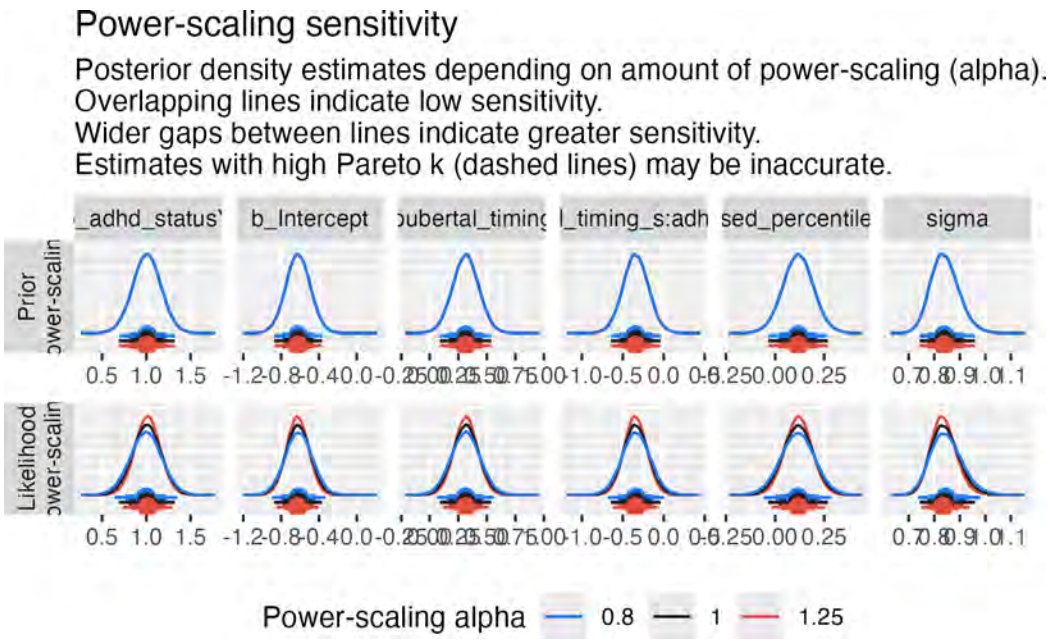
```
                              b_pubertal_timing_s 0.008      0.072        -
                                   b_adhd_statusY 0.022      0.098        -
                                b_sed_percentile_s 0.004      0.092        -
b_pubertal_timing_s:adhd_statusY 0.011      0.080        -
                                            sigma 0.000      0.139        -
                                        Intercept 0.000      0.079        -
```

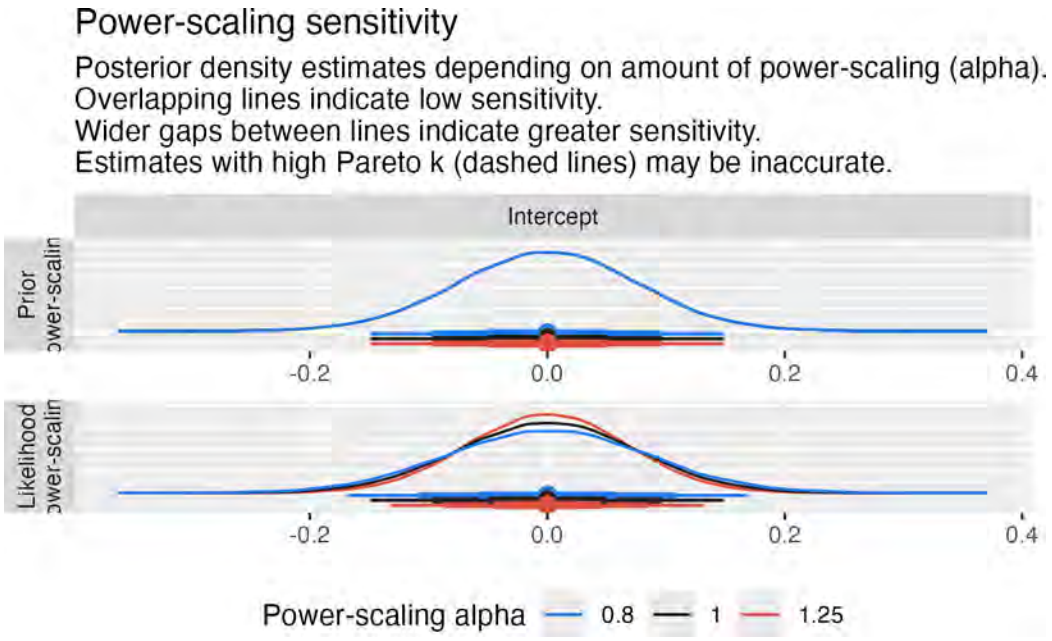




```
Sensitivity based on cjs_dist
Prior selection: intercept
Likelihood selection: all data
```

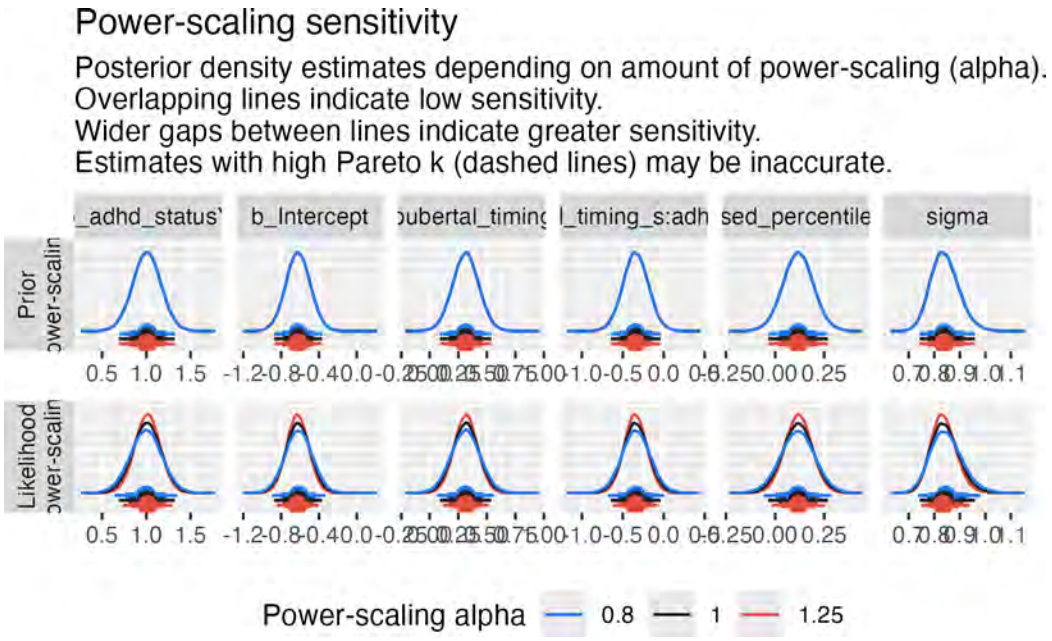
variable	prior	likelihood	diagnosis
b_Intercept	0	0.096	-
b_pubertal_timing_s	0	0.072	-
b_adhd_statusY	0	0.098	-
b_sed_percentile_s	0	0.092	-
b_pubertal_timing_s:adhd_statusY	0	0.080	-
sigma	0	0.139	-
Intercept	0	0.079	-

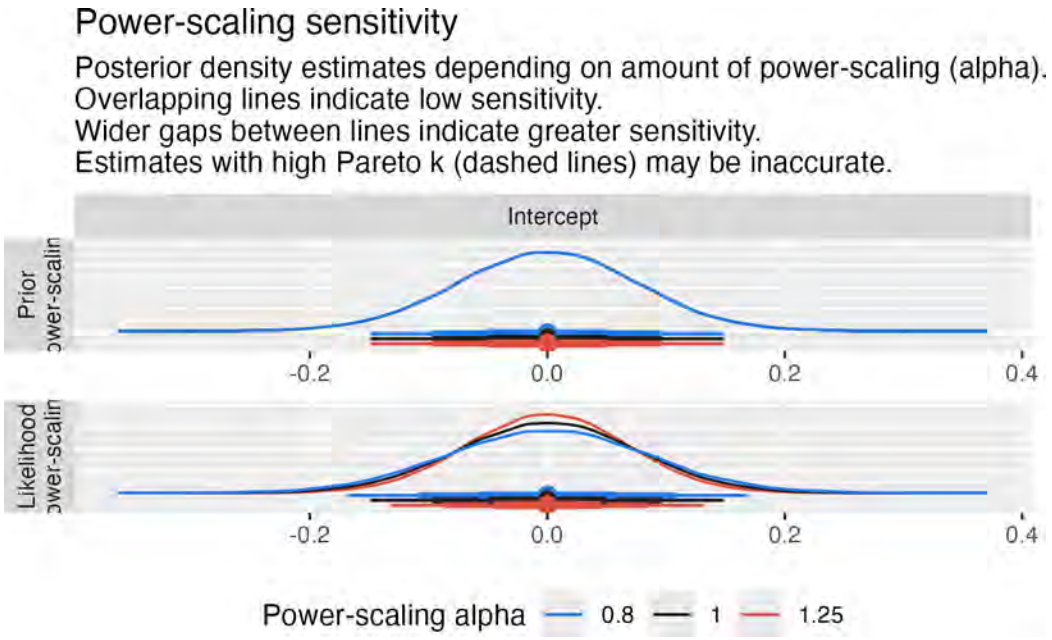





```
Sensitivity based on cjs_dist
Prior selection: sigma
Likelihood selection: all data
```

variable	prior	likelihood	diagnosis
b_Intercept	0.001	0.096	-
b_pubertal_timing_s	0.001	0.072	-
b_adhd_statusY	0.001	0.098	-
b_sed_percentile_s	0.001	0.092	-
b_pubertal_timing_s:adhd_statusY	0.001	0.080	-
sigma	0.009	0.139	-
Intercept	0.001	0.079	-





H2 Model interaction

```
=== Simple slopes analysis ===
```

Group	Slope (Mean)	95% CI	pd	% in ROPE
N	0.31	[0.09, 0.54]	99.7%	3%
Y	-0.03	[-0.23, 0.18]	60.9%	64.9%

```
=== Pubertal timing effect on emotional reactivity (raw) in non-ADHD group ===

# A tibble: 1 x 3
  slope_mean_raw lower_95_raw upper_95_raw
           <dbl>        <dbl>        <dbl>
1           2.95        0.847         5.06
```

H2 Model interaction graph

